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MILITARY SPECIFICATION

CAMERA, STILL PICTURE, PANORAMIC, KA-99A

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the KA-99A type of 5-inch (127mm) panoramic camera, herein referred to as the camera.

2. APPLICABLE DOCUMENTS

2.1 Governnent documents

2.1.1 Specification, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

PPP-B-621	Boxes, Wood, Nailed and Lock Corner.

MILITARY

MIL-T-27	Transformer and Inducter (Audio, Power, and High Power Pulse, General Specification for.
MIL-P-116	Preservation-Packing, Methods of.
MIL-C-675	Coating of Glass Optical Elements (Anti-Reflection).
DOD-D-1000	Drawing, Engineering and Associated List.
MIL-W-5088	Wiring, Aerospace Vehicle.
MIL-E-5400	Electronic Equipment, Aerospace, General Specification for.
MIL-E-5558	Enamel, Wrinkle-Finish for Aircraft Use.
MIL-M-7793	Meter, Time Totalizing.
MIL-M-9868	Microfilming of Engineering Documents, 35mm Requirements for.
MIL-M-13508	Mirror Front Surface Aluminized for Optical Elements.
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment, Including Ground Support Equipment.
MIL-C-24308	Connectors, Electric, Rectangular, Miniature Pola ed Shell, Rack and Panel, General Specification for.
MIL-M-38510	Microcircuits, General Specification for.
MIL-C-39006	Capacitor, Fixed, Electrolytic (Nonsolid Electrolyte) Tantalum, Established Reliability, General Specification for.

NAVAL AIR SYSTEMS COMMAND

AS-4613	Application and Derating Requirements for Electronic Components, General Specification for.

STANDARDS

FEDERAL

FED-STD-H28	Screw-Thread Standards for Federal Services.
FED-STD-595	Colors.

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-150	Photographic Lenses.
MIL-STD-155	Joint Photographic Type Designation System.
MIL-STD-415	Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for.

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MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-461	Electromagnetic Interference Characteristic Requirements for Equipment.
MIL-STD-462	Electromagnetic Interference Characteristics Measurement of.
MIL-STD-470	Maintainability Program Requirements (For Systems and Equipments).
MIL-STD-471	Maintainability Demonstration.
MIL-STD-680	Contractor Standardization Program Requirements.
MIL-STD-681	Identification Coding and Application of Hook-up and Lead Wire.
MIL-STD-704	Aircraft Electrical Power Characteristics.
MIL-STD-781	Reliability Tests, Exponential Distribution.
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production.
MIL-STD-810	Environmental Test Methods.
MIL-STD-882	System Safety Program Requirements.
MIL-STD-883	Test Methods and Procedures for Microelectronics.
MIL-STD-965	Parts Control Program.
MIL-STD-1278	Filters, Light, Photographic.
MIL-STD-1353	Electrical Connectors, Plug-in Sockets and Associated Hardware, Selection and Use of.
MIL-STD-1390	Level of Repair.
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1562	Lists of Standard Mirocircuits.
MIL-STD-1629	Procedure for Performing a Failure Mode, Effects and Criticality Analysis.
MIL-STD-2077	Test Program Sets, General Requirements for.
MIL-STD-2084	Maintainability of Avionic and Electronic Systems, General Requirements for.
MS-26565	Spool, Photographic Film Aerial.

HANDBOOKS

MILITARY

MIL-HDBK-217	Reliability Stress and Failure Rate Data for Electronic Equipment.
MIL-HDBK-472	Maintainability Prediction.

2.1.2 Other government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL AVIONICS CENTER

NAC 1213AS140	Installation, Camera, KA-99A.

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PUBLICATIONS

NAVAL AIR SYSTEMS COMMAND

NAVAIR 16-1-525 Preferred Standard Avionics Test Equipment.

NAVAIR 19-1-127 Preferred List of Common Ground Support Equipment (Nonavionic).

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issue of the document which is indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARD INSTITUTE

PH1.10-1976 Dimensions for Unperforated and Perforated Photographic Film in Rolls, Including Leaders and Trailer, for Aerial and Related Uses.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.5 and 6.3).

3.2 Parts, materials, and processes. The selection and application of parts, materials, and processes shall be in accordance with MIL-E-5400 except as specified herein. Only electronic components that are interchangeable and can be replaced with no design changes shall be incorporated in the production camera to meet the derating requirements specified herein.

3.2.1 Screening of parts. All electrical, electronic and electromechanical parts shall be subjected to 100 percent screening. Nonstandard parts and non-established reliability military standard parts shall be screened to the requirements of the most similar military standar or standard-established reliability part, as well as the requirements of TABLE I. As a minimum, all integrated circuits shall be screened to MIL-STD-883, Method 5004, Class B. As a minimum, hybrid circuits shall be screened to MIL-STD-883, Method 5008.

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cellophane shall be long enough to fold back and cover the ... One sheet of the

3.2.2 Parts derating and application. Application and derating requirements shall be in accordance with AS-4613.

3.2.2.1 Application limitations. Limitations on parts usage shall be as specified in 3.2.2.1.1 and 3.2.2.1.2.

3.2.2.1.1 Prohibited application. The use of electron tubes, aluminum electrolytic capacitors, fixed paper and plastic dielectric capacitors in molded cases, plastic epoxy or other encapsulated semiconductors and germanium semiconductors shall be prohibited.

3.2.2.1.2 Restricted application. The use of non-solid sintered-slug tantalum capacitors is restricted to MIL-C-39006, CLR 79, qualified devices. Programmable Read Only Memories (PROMs) and hot carrier (Schottky) rectifiers shall not be used in production equipment.

3.2.3 Stress analysis. The contractor shall conduct a part-level stress analysis to verify derating compliance. This analysis shall verify that all parameter deratings are compliant under the conditions of nominal unit and module power dissipation, worst case piece part stress, and worst case equipment thermal environment, including worse case altitude and temperature. The contractor shall prepare a reliability stress analysis report in accordance with MIL-STD-785 (see 6.2.2).

3.2.4 Parts control and standardization program. The contractor shall establish and maintain a parts selection, parts control, and parts standardization program in accordance with MIL-STD-965, Procedure I, and MIL-E-5400. The contractor shall prepare a standardization program plan in accordance with MIL-STD-680 (see 6.2.2). The program shall be applicable to any equipment designed or modified by the contractor or his subcontractors, but shall not apply to unmodified off-the-shelf equipment.

3.2.5 Nonstandard part and material approval. Approval for the use of nonstandard parts and materials, including electron tubes, transistors, and diodes, other than microelectronic devices, shall be obtained as specified in MIL-E-5400 and MIL-STD-965, Procedure I. The contractor shall prepare requests for approval of nonstandard parts and to proposed additions to an approved Program Parts Selection List (PPSL) (see 6.2.2). Microelectronic devices shall be approved as specified in MIL-STD-965, except that the dual source and interchangeability requirement shall not apply. Approval as a nonstandard part shall not be required for devices which meet the criteria of being listed in MIL-STD-1562, available from at least two sources which are listed in the QPL of MIL-M-38510, and procured to the requirements of MIL-M-38510 from a qualified source.

3.2.5.1 Samples for parts and materials approval. The sample requirements for nonstandard parts and materials shall be in accordance with MIL-E-5400.

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TABLE I. Supplemental part screening. 1/

100% Environmental Screen						100% Burn-in	
Part	Temp. Cycling	High Temp.	Case Leakage	Centri-fuge	Other	Time Temp.	Parameters
Paper and/or plastic dielectric caps	-85°F(-65°C) 257°F(125°C) 10 cycles					100±5hr. at rated voltage and max. rated temp.	C, Insul. resistance DF, Dielectric with standing voltage
Fixed metalized paper dielectric D.C.	-76°F(-60°C) 257°F(125°C) 10 cycles					100±5hr. at rated voltage at max. ambient temp.	C, Insul. resistance DF, Dielectric with standing voltage
Feed thru RF det AC & DC	-85°F(-65°C) 257°F(125°C) 10 cycles					100±5 hr. at rated voltage at max. ambient temp.	C, Insul. resistance DF, Dielectric with standing voltage
Resistors, carbon comp. and others.	-85°F(-65°C) 257°F(125°C) 10 cycles				2/		
Diodes	-85°F(-65°C) 392°F(200°C) 10 cycles	392°F (200°C)	Pressure bomb	20KG	Power Pulse (Pre-Cap) Visual	160 hr. at max. rated power at 257°F (125°C)	Group A of nearest military establi: part.
Transistors	-85°F(-65°C) 392°F(200°C) 10 cycles	392°F (200°C)	$5x10^{-8}$ cc/sec	20KG	Rev Bias 302°F (150°C) (PNP) (Pre-Cap) Visual	168 hr. at max. rated power at 257°F(125°C) HTRB PHP only	Group A of nearest military established part.

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TABLE I. Supplemental part screening. - Continued

100% Environmental Screen						100% Burn-in	
Part	Temp. Cycling	High Temp.	Case Leakage		Other	Time Temp.	Parameters
Integrated Circuits					MIL-STD-883 Method 5004,		
Hybrids					Class B MIL-STD-883 Method 5008		
Transformers and Inductors	-67°F(-55°C) 257°F(125°C) 10 cycles						Group A inspection per MIL-T-27

1/ This table only applies where the above requirements are not called out in the closest military established reliability specification.

2/ Parts to be manufactured on established reliability line or, burn-in of 100 hrs with max. power at 77°F(25°C).

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3.2.6 Microelectronic modular assemblies. When used, microelectronic modular assemblies shall meet the requirements of MIL-STD-965. Conformal coatings, encapsulants, embedments, or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly.

3.2.7 Modules. The electronic portions of the camera shall be modularized to the greatest extent possible in accordance with MIL-STD-2084.

3.2.8 Non-repairable subassemblies. When used, non-repairable subassemblies shall be in accordance with MIL-E-5400 and MIL-STD-2084.

3.2.9 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other.

3.2.10 Screw threads. Screw threads used in the construction of the camera shall be in accordance with FED-STD-H28.

3.2.11 Locking devices and lock washers. Screw threaded assemblies subject to loosening by vibration shall be provided with locking devices or lock washers in accordance with MIL-E-5400.

3.2.12 Protective treatments. When materials are used in the construction of the camera that are subject to deterioration when exposed to environmental conditions likely to occur during service usage, the materials shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements specified herein. The use of any protective coating that will crack, chip, or scale with age or extremes of environmental conditions shall be prohibited.

3.3 Design. The camera shall be designed to be mounted in the Tactical Air Reconnaissance Pod System (TARPS) utilizing the KA-99A cradle assembly, Naval Avionics Center Drawing 1213AS140. The mounting locations and center of gravity of the camera shall be in accordance with FIGURE 1. The camera shall be designed to operate in conjunction with the AN/ASQ-172A Data Display Set, and the C-10491 Control Indicator-Power Distribution Unit. The camera design shall be in accordance with the applicable requirements of MIL-E-5400 for design.

3.3.1 Weapon replaceable assemblies (WRAs). The camera shall be modular in design. The camera shall be separated into two WRAs. The camera WRAs shall be the camera body and magazine assembly (with supply and take-up cassettes installed) and the camera electronics assembly. The magazine assembly shall be capable of being removed from the camera body for the purpose of performing scheduled and unscheduled maintenance. The WRA design shall be such that any WRA, Shop Replaceable Assembly (SRA), or sub-SRA may be functionally inspected at the intermediate maintenance level. All necessary adjustments, alignments, and calibrations shall be accomplished at the intermediate maintenance level to eliminate the need

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for adjustments, alignment, and calibration at the organization maintenance level. Suitable test points shall be incorporated into the WRA design to eliminate the need for extender devices, such as extender cards, cables, etc., at all maintenance levels. The WRA design shall include the requirements for modularization and standardization as specified in MIL-STD-2084. The WRA design shall be such that fail detection and isolation shall be accomplished at the intermediate maintenance level without the need to use maintenance-assist modules during repair operations. Definitions for WRA, SRA, and Sub-SRA shall be in accordance with MIL-STD-2084.

3.3.2 Built-in-test (BIT). BIT provisions shall be designed to detect major voltage levels, timing, phase, input/output tolerance requirements, and film jamming. The BIT shall test the scan servo, Automatic Exposure Control (AEC), range servo, and Forward Motion Compensation (FMC). A camera failure or degraded condition detected by by either automatic or manual BIT initiation shall be indicated by the use of latching indicators that shall be visible to maintenance personnel upon gaining access to the camera compartment. The design of BIT shall be such that initiation of BIT can be acco plished from either the camera or from a remote location by appropriate signal input as specified in 3.9.1.1.5. A failure or degraded condition in the camera shall be indicated by a 28VDC output signal for each indicator at one of the camera external connectors. The 28VDC signal shall be capable of supporting a 500 milliamperes (ma) load.

3.3.2.1 BIT features. BIT features shall eliminate the need for any auxiliary test devices or Peculiar Ground Support Equipment (PGSE) at the organizational level for the performance of WRA fault detection or fault isolation. BIT provisions shall be added to an item for the sole purpose of inspecting the item. The provisions, or circuitry, shall be simple in design and operation, accurate to within plus or minus two percent of the measured parameter, easily maintained, and shall not degrade the performance of the item in which the circuitry is incorporated. Failure of the BIT circuitry shall not cause a failure of the camera with which it is associated. BIT provisions shall provide optimum convenience of use and operation. The design of controls and readout devices shall be such that the controls and readout devices can be easily used and interpreted by low-skill military personnel at the E-4 level. BIT provisions shall be designed to automatically provide fault information in a manner which will minimize the need for referrals to instruction and maintenance handbooks and through the self-check feature, improve the camera readiness (availability).

3.3.2.2 Self-test cycle. The BIT design shall include a self-test cycle to operate the camera for a maximum of 15 seconds to test internal functions such as film transport, scan servo operation, and AEC servo operation. The design of the self-test cycle shall be such that the self-test cycle will be activated as a part of the camera BIT sequence. Failures or degraded condition detected during the operation of the self-test cycle will be activated as a part of the camera BIT sequence.

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Failures or degraded conditions detected during the operation of the self-test cycle shall be automatically isolated to a single WRA, and shall be indicated by activation of the appropriate latching indicator as specified herein.

3.3.3 Test points and test features. Suitable test points and test features shall be included in the design of the camera. Selection and accessibility of test points shall be compatible with the design of replaceable assemblies and the level of maintenance at which the test points are to be used.

3.3.4 Safety. The design of the camera shall incorporate the personnel safety requirements of MIL-STD-454, Requirement 1. The contractor shall prepare a safety statement, a system safety program plan, and a hazard analysis report in accordance with MIL-STD-882 (see 6.2.2)

3.3.5 Human engineering. The applicable principles and criteria specified in MIL-STD-1472 shall be incorporated into the design of the camera.

3.4 Reliability. The reliability requirements for the camera shall be 50,000 Mean Cycles Between Failures (MCBF) for the electromechanical portions (scanning slit mechanism and film transport) of the camera. The electronic functions and optical elements of the camera shall have a Mean Time Between Failure (MTBF) of 2,000 hours. The total camera shall have a MTBF of 117 hours. The conversion of MCBF to MTBF shall be based on mission duration time of 2 hours and 2,000 feet (609.6m) of film expended per mission.

3.4.1 Reliability program. The contractor shall establish and conduct a reliability program for the development and inspection of the camera in accordance with MIL-STD-785 and MIL-STD-781.

3.4.1.1 Reliability program plan. The contractor shall prepare a reliability program plan (production) for the camera in accordance with MIL-STD-785 (see 6.2.2).

3.4.1.2 Reliability analysis and predictions. The contractor shall prepare a reliability analysis and predictions report of the camera in accordance with MIL-STD-785 and MIL-HDBK-217 (see 6.2.2).

3.4.1.3 Failure modes, effects and criticality analysis (FMECA). The contractor shall prepare a FMECA report of the camera in accordance with MIL-STD-785 and MIL-STD-1629 (see 6.2.2).

3.4.1.4 Failure summary and analysis. The contractor shall prepare a failure summary and analysis report of the camera in accordance with MIL-STD-785 and MIL-STD-781 (see 6.2.2).

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3.4.2 Operational stability. The camera shall operate as specified, continuously or intermittently, for a period of at least 100 hours without the necessity for readjustment of any controls which are inaccessible to the operator.

3.4.3 Service life. The camera shall have a service life of at least 1,500,000 cycles when operated under any combination of environmental conditions specified herein. Allowance shall be made for corrective maintenance by replacement of worn or damaged parts.

3.5 Construction. The camera shall be constructed in such a manner that the physical and functional relationships between camera components are not affected by the strains, impacts, and vibration associated with operation, storage, and shipping. The construction of the camera shall be compatible with space limitations and shall be capable of withstanding without damage a Class 2 flight environment as specified in MIL-E-5400.

3.5.1 Strength and rigidity of alignment. Each part of the camera shall be of sufficient strength and rigidity to accomplish its respective function and to maintain smooth operation with proper fit and precise alignment throughout the life of the camera. All parts of the drive mechanism that require timing shall be provided with timing marks.

3.5.2 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with respect to installation, form, fit, function, and performance as specified in MIL-E-5400. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of DOD-D-1000.

3.5.3 Lighttightness. The camera shall be constructed to be lighttight as specified herein. The camera magazine assembly shall be lighttight to prevent the fog exposure of the film on the spool within the cassette when the magazine assembly has been removed from the camera body. The film cassette shall be lighttight to the extent that when loaded with 2,000 feet 609.6m) of film having an Aerial Film Speed (AFS) of 200, no mo than the first four wraps of film shall be capable of being fog exposed.

3.5.4 Fabrication. Machining, drilling, and forming shall be accomplished with the use of accurate templates, jigs and gauges. Boxes, cases, shields, and compartments shall be fabricated by casting, drawing o bending and welding, except where ease of servicing of the camera requires that a removable panel construction be used. Where applicable stresses dictate the use of a strong aluminum or magnesium alloy that does not provide a good weld, riveting or bolting may be used.

3.6 Maintainability. The camera shall have a minimum number of parts consistent with reliability and shall be contained in a single assembly or minimum number of assemblies. The design shall utilize readily removable and replaceable units to facilitate easy assembly, disassembly,

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location of trouble sources and repair, with a minimum "down time" of the complete camera.

3.6.1 Mean-time-to-repair (MTTR). The MTTR at the organizational level and the intermediate level shall be as specified herein.

3.6.1.1 Organizational level. The MTTR of the camera at the organizational level shall be not greater than 15 minutes. The maximum time-to-repair shall be not greater than 45 minutes for 95 percent of all corrective maintenance actions. The specified MTTR shall include the time to verify the fault/failure, the time to isolate to the faulty WRA, the time to remove and replace the faulty item, and the time required to verify the effectiveness of the repair. The time required to obtain spare parts and facilities as well as access time shall not be included. The camera shall be designed so that the alignment, adjustment, and calibration of the WRA will not be required upon replacement at this maintenance level.

3.6.1.1.1 Detection and isolation. The camera BIT shall be capable of detecting at least 80 percent of the camera critical failures at the organizational level. A failure shall be classified critical if it results in a mission abort, no imagery of unuseable imagery. At least 90 percent of the detected critical failures shall be isolated to a single faulty WRA at the organizational level and at least 99 percent of the failure indications shall result from actual camera failures.

3.6.1.2 Intermediate level. The MTTR of the camera at the intermediate level shall be not greater than one hour. Maximum time to repair at the intermediate level shall be not greater than 2.5 hours for 95 percent of all corrective maintenance actions. The specified MTTR shall include the time to perform confirming inspections even though the removed item was identified as faulty during organizational maintenance. Alignment, adjustment, and calibration time shall be included. Time awaiting replacement parts or facilities shall not be included.

3.6.2 Test points and facilities. Test points and facilities shall be as specified in 3.6.2.1 and 3.6.2.2.

3.6.2.1 Electronic test points. Electronic test points shall be in accordance with MIL-STD-2084 and MIL-STD-415. When in conflict, MIL-STD-2084 shall govern. All test points shall be designed to prevent overloading of the circuit under test and shall be designed so that damage to the unit shall not result from a short circuit between test points and any test point ground. Test points shall be conveniently located and shall be accessible without removal of any circuit card or component, except where that test is utilized for component isolation on the circuit card. All test points shall be legibly marked.

3.6.2.2 Mechanical test points. Mechanical test points shall be provided as required and shall be in accordance with MIL-STD-415. Mechanical test points shall be simple in design and shall require minimal use of special tools.

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3.6.3 Lubrication. Wherever practicable, the camera shall utilize permanently lubricated sealed bearings and shall be designed to minimize requirements for routine lubrication.

3.6.4 Maintenance concept. The design of the camera shall be consistent with the maintenance concept specified herein. The camera shall b designed to be repaired at the organizational maintenance level by removal and replacement of the WRAs without the use of external test equipment. T camera shall be designed to be repaired to the bit and piece part level at the intermediate maintenance level.

3.6.4.1 Maintenance skill levels. The camera maintainability design shall permit the optical/mechanical maintenance tasks to be performed by a Photographer's Mate and the electronic equipment maintenance tasks to be performed by an Aviation Electronics Technician, each with the respective skill levels specified herein.

3.6.4.1.1 Organizational maintenance skill levels. Organizational maintenance shall be performed by personnel possessing skill levels of E-4 rank and above having successfully completed the appropriate military bran service and technical schooling.

3.6.4.1.2 Intermediate maintenance skill levels. Intermediate maintenance shall be performed by personnel possessing skill levels of E-5 rank and above having successfully completed the appropriate military bran service and technical schooling.

3.6.4.2 Organizational level maintenance. Organizational level maintenance tasks shall be as specified in 3.6.4.2.1 and 3.6.4.2.2.

3.6.4.2.1 Scheduled Maintenance. Scheduled maintenance shall be limited to the removal of exposed film, the loading of unexposed film, the setting of the preflight exposure compensation (Effective Aerial Film Speed/ Filter Factor Control (S/C), visual inspection of optical surfaces and film transport elements for foreign matter and cleaning when required, installation and removal of lens filters and protective cover, and performing preflight inspection by activation of BIT. No scheduled maintenance action at this level shall be greater than 10 minutes.

3.6.4.2.2 Unscheduled maintenance. Unscheduled maintenance shall consi of fault detection and fault isolation to a single WRA using only BIT and removal and replacement of the faulty WRA and verification of corrective action by activation of BIT. To the maximum extent possible, the camera shall be designed so that removal and replacement of a faulty WRA can be accomplished without removing additional WRAs while the camera is installe in the Tactical Air Reconnaissance Pod System (TARPS). No fault detection isolation PGSE will be used at this maintenance level. Upon replacement o the WRA, no adjustment, alignment, or calibration shall be required to ada the WRA to other camera WRAs or other camera subsystems.

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3.6.4.3 Intermediate level maintenance. Intermediate level mainten e tasks shall be as specified in 3.6.4.3.1 and 3.6.4.3.2.

3.6.4.3.1 Scheduled maintenance. Scheduled maintenance shall include t loading of film in the magazine assembly and the periodic inspection, cleaning and lubrication of the camera.

3.6.4.3.2 Unscheduled maintenance. Unscheduled maintenance shall consist of checkout, fault detection, and fault isolation of electronic and optical mechanical WRAs, repair of the WRAs by replacement of SRAs, and verification of Ready For Issue (RFI) condition. Repair of faulty SRAs and subassemblies will consist of fault detection, fault isolation, removal and replacement of faulty piece parts, and verification of RFI condition. Alignment and adjustment of WRAs at this maintenance level shall eliminate the need for any further alignment and adjustment when the WRA is mated with other camera components. The camera shall be Automatic Test Equipment (ATE) compatible. Any requirements for general purpose or common electronic test equipment or other Ground Support Equipment (GSE) shall be selected from NAVAIR 16-1-525 and NAVAIR 19-1-127 . PGSE shall be limited to adapter and interconnecting devices required to adapt the camera WRAs and SRAs to standard equipment and ATE, unless otherwise authorized by the procuring activity. All interconnecting devices shall be designed in accordance with MIL-STD-2077.

3.6.4.4 Depot level maintenance. Depot level maintenance shall consist of the repair of electrical, electronic, and optical/mechanical compo ts that are beyond the capability of intermediate level maintenance.

3.6.4.5 Level of repair (LOR). The contractor shall prepare an LOR program plan, an LOR analysis report, and an LOR summary report of the camera in accordance with MIL-STD-1390 (see 6.2.2).

3.6.4.6 Extender devices. Extender devices, such as cards and adapters, shall not be used to perform normal maintenance and fault isolation at any maintenance level.

3.6.5 Maintainability program. The contractor shall establish and maintain a maintainability program for the camera in accordance with MIL-STD-470. The maintainability program shall assure that all maintenanc tasks shall be capable of being performed by technicians within the comprehensive skill levels and maintenance times specified in 3.6.5.1 through 3.6.5.3.

3.6.5.1 Maintainability program plan. The contractor shall prepare a maintainability program plan for the camera in accordance with MIL-STD-470 (see 6.2.2).

3.6.5.2 Maintainability analysis and predictions. The contractor shall prepare a maintainability analysis and predictions report of the camera in accordance with MIL-STD-470 (see 6.2.2). The prediction technique shall b

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in accordance with MIL-HDBK-472, Procedure II, Part A, for electronic equipment and Part B, converting man-hour rate to time-to-repair, for mechanical/optical components for corrective maintenance.

3.6.5.3 Commercial equipments. The maintainability of all commercial assemblies constructed and purchased as off-the-shelf equipment shall be included in the maintainability program and shall meet the maintainability requirements specified herein.

3.7 Detail of components. The camera shall consist of the following major assemblies:

Description	Quantity
a. Camera body	1 each
b. Camera electronics Unit	1 each
c. Magazine assembly	1 each
d. Cassette	2 each

3.7.1 Camera body. The camera body shall contain the optical barrel, camera timing mechanism, scan drive, magazine assembly, FMC mechanism, AEC mechanism, solid-state recording head, and camera service life counter. The camera body shall include provisions for mounting the camera electronics unit on the camera body or in a remote location when required by the TARPS. The camera body shall provide a film format in accordance with FIGURE 2. The camera body shall include the wiring required to inspect the camera body and the magazine assembly as a unit.

3.7.1.1 Optical barrel. Imagery is obtained by rotation of the optical barrel which includes the scan mirror, lens, and focal plane mirror. The optical barrel shall be synchronized with the camera timing mechanism. The optical barrel shall make one revolution for each photographic cycle.

3.7.1.2 Scan and film transport drive servo. The scan and film transport drive servo shall drive the optical barrel assembly, the film transport mechanism, and shall control the rate of exposure across the scanning slit. The scan rate shall be proportional to the ground velocity divided by altitude above ground level (Vg/H) input signal during autocycle operation. The servo shall provide accurate speed control over the cycle range of 0.38 to 8 cycles per second (cps). The servo shall provide sufficient torque to accelerate the camera to eight cps within the first five seconds of camera operation. Under no circumstances shall more than 20 frames of film be expended during the initial five-second start up period.

3.7.1.3 Film transport mechanism. The film transport mechanism shall consist of suitable drive rollers, guides, sprockets, and film keepers required to properly transport a 2,000 ft (609.6m) roll of 5-inch (127mm) film having a nominal base thickness of 2.5 mils and Type II

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perforations in accordance with ANSI PH1.10. The film transport mechanism shall also be capable of properly transporting 5-inch (127mm) aerial film with a nominal base thickness of 4.0 mils and Type II perforations in accordance with ANSI PH1.10. The film transport mechanism shall not generate static marks which would degrade the imagery on the processed film. The film transport mechanism shall not damage the film in any manner.

3.7.1.4 Forward motion compensation mechanism. When the camera is operating in the autocycle mode, as specified in 3.9.2.1, FMC shall be accomplished by the FMC mechanism during the exposure in the appropriate direction as dictated by the design configuration of the optical path, location of the focal plane, and the direction of flight. The FMC mechanism shall be disengaged in the air-to-air mode of operation. The FMC mechanism shall automatically disengage during the pulse mode of operation. The disengagement shall not necessarily coincide with the pulse mode switch point.

3.7.1.4.1 FMC rates. The camera shall provide FMC rates over the autocycle operating range which shall correspond to the Vg/H range of 0.08 to 1.79 radians per second (0.049 to 1.06 knots per foot).

3.7.1.5 Automatic exposure control mechanism. The camera shall be provided with a servo system capable of accepting a Vg/H ratio signal and appropriate light sensor signal to automatically position the shutter slit width and lens diaphragm opening to provide for proper exposure of film having AFS throughout the S/C range of 10 to 1200. The S/C settings shall be 10, 20, 40, 60, 80, 100, 120, 200, 400, 600, and 1200. The AEC mechanism shall provide automatic exposure control for all modes of camera operation throughout a scene brightness range of 12 to 8000 foot-lamberts with a Wratten-12, or equal, filter installed over the lens and when compatible with camera cycle rate and AFS rating of the film. Under conditions of decreasing luminance, the AEC mechanism shall increase the aperture of the iris diaphragm to maximum before increasing the shutter slit width. Under conditions of increasing luminance, the AEC mechanism shall decrease the shutter slit width to its minimum before decreasing the aperture of the iris diaphragm. When the camera is operating in the port or starboard standoff mode, the AEC mechanism shall provide proper exposure on the film for the selected scan area.

3.7.1.5.1 Frame exposure accuracy. The exposure for any frame shall be within plus or minus one-half stop of the correct value, except when the port or starboard standoff modes are selected. Only that portion of the exposed film corresponding to the standoff mode selected shall be within plus or minus one-half stop of the correct exposure value.

3.7.1.5.2 Exposure override. Provisions shall be made for exposure override. This function shall interface with the exposure override control system in the aircraft specified in the contract.

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3.7.1.6 Camera data recording. The camera shall be capable of recording on film Binary Coded Decimal (BCD) data matrix, alphanumeric data block, numeric frame count, center of format fiducial marks, direction of film travel mark, and flight direction mark. The BCD data matrix and the alphanumeric data block require inputs from the AN/ASQ-172A Data Display Set installed in the TARPS. The center of format fiducial marks, direction of film travel mark, and flight direction mark shall be considered fixed data. The direction of film travel mark and the flight direction mark shall be considered fixed data. The direction of film travel mark and the flight direction mark shall be combined into the single symbol, "⌊", as specified in 3.7.1.6.4. Time sequencing for the numeric frame count, center of format fiducial marks, and film direction/flight direction mark shall be in accordance with FIGURE 2.

3.7.1.6.1 Solid-state recording head. The camera shall be provided with a solid-state recording head. The recording head shall be capable of being operated by the AN/ASQ-172A Data Display Set. The recording head shall be capable of being removed from and replaced into the camer The recording head shall be capable of recording a BCD data matrix in accordance with FIGURE 3. The location of the data block recorded o the film shall be in accordance with FIGURE 2. The recording head shal be capable of recording an alphanumeric data block in accordance with FIGURE 4 and as specified herein. The recorded data block shall consis of eight rows each having eight characters. Each character shall be formed by a five dot by seven dot matrix. The dimensions and location of the alphanumeric data block recorded on the film shall be in accordance with FIGURE 2.

3.7.1.6.2 Numeric frame count. The camera shall be provided with a three-digit resettable counter to record on the film the frame count fo each frame of film as it is exposed during camera operation. The dimensions and location of the frame count shall be in accordance with FIGURE 2. The counter shall be reset to zero upon application of aircraft power (28VDC) to the camera. The counter shall be reset to zero upon any interruption of aircraft power to the camera that exceeds one minute.

3.7.1.6.3 Fiducial marks. The camera shall be equipped with two fiducial marks. The fiducial marks shall indicate the center of the film format along the scan direction (90 degrees). The fiducial marks shall be recorded along the top and bottom edges of the film in accordance with FIGURE 2.

3.7.1.6.4 Flight direction and direction of film travel marks. The camera shall record a flight direction mark and a direction of film travel mark between each frame of film. The two marks shall be combine into the single symbol, "⌊", and shall be recorded on the film in accordance with FIGURE 2.

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3.7.1.7 Camera cycle counter. The camera body shall contain a seven digit, additive, and nonresettable counter. The counter shall record the total number of camera cycles. The counter shall be located in a visible location on the exterior of the camera body.

3.7.1.8 Flight direction indicator. A permanently legible arrow with the words, "direction of flight", shall be attached to a visible location on the exterior surface of the camera to indicate the direction of flight when the camera is mounted in the TARPS.

3.7.1.9 Lens. The camera shall be provided with a 9-inch (228.6mm) plus or minus 0.136-inch (3.5mm) focal length, f/4.0 lens with sufficient field of view to prevent vignetting of the film during all modes of camera operation. The lens shall be a Type I lens in accordance with MIL-STD-150, suitable for aerial reconnaissance. The lens shall properly mount in the optical barrel.

3.7.1.9.1 Antireflection coating. All air-to-glass surfaces shall have reflection reducing coatings in accordance with MIL-C-675.

3.7.1.9.2 Lens diaphragm. The lens shall be provided with a between the lens iris diaphragm. The diaphragm shall be capable of being automatically varied by the AEC mechanism. The diaphragm shall be calibrated from f/4.0 to f/16.0 in accordance with MIL-STD-150.

3.7.1.9.3 Lens filters. The camera shall be provided with three glass filters, Wratten-12, or equal, Wrattn-25, or equal, and clear. The filters shall be capable of being installed directly in front of the camera lens through the scan mirror viewport in the optical barrel without the use of special tools.

3.7.1.10 Mirrors. Mirrors shall be fabricated of an ultra-low expansion material, such as CER-VIT, or equal. Reflecting surface shall be polished to a flatness consistent with system optical performance requirements. The mirrors shall be aluminized and overcoated with protective material in accordance with the requirements of MIL-M-13508.

3.7.1.11 Optical transmittance. The camera optical system shall have a T stop value not greater than T-5.

3.7.1.12 Optical field of view. The camera optical system shall provide a 27 degree 20 minute longitudinal angle of view and shall be capable of a 180 degree lateral scan angle. The optical system shall provide sufficient coverage to allow for full range FMC displacement of the lens. Vignetting of the optical bundle shall not occur within the specified optical field of view during any mode of camera operation, except that vignetting of approximately 50 percent of the light bundle at 0 degree and 180 degrees of lateral scan shall be permitted. The vignetting shall decrease to 0 percent in transition regions of approximately 7 degrees at start and end of lateral scan.

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3.7.1.13 Electronics and electromechanical devices. The camera body shall be provided with a power supply, electronics circuitry, and electro-mechanical devices to perform Vg/H signal conditioning, timing and sequencing, scan and film drive servo, AEC compensation, including aperture and slit width positioning, range focus positioning, data recording, and fiducial mark imaging.

3.7.2 Camera electronics unit. The camera electronics unit shall interface the camera with the TARPS power and control signals. The camera electronics unit shall provide the signals required for the numeric frame count, center of format fiducial marks, direction of film travel mark, and flight direction mark. The camera electronics unit shall contain the electronics circuitry and controls for the camera. The camera electronics unit shall contain a time totalizing meter in accordance with MIL-M-7793. The S/C switch, BIT initiate switch, BIT indicators, and test points (via test connector) shall be located in the camera electronics unit. All of the control devices shall be modularized to the greatest extent possible. Removable circuit boards shall be utilized wherever possible. The camera electronics unit shall be capable of being mounted on the camera body or remotely as required by the TARPS.

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3.7.3 Magazine assembly. The camera shall be provided with a film magazine as a part of the magazine assembly. The film magazine shall contain the film transport mechanism. The magazine shall provide the necessary seating surfaces and fasteners for mounting the two cassettes to the magazine and for mounting the magazine to the camera body. The magazine shall be of sufficient strength and rigidity to maintain proper alignment throughout the life of the camera. Mechanical and electrical connections required between the magazine and the camera body shall be automatically engaged or disengaged when the components are mated or separated, respectively. Mechanical connections shall include all mechanical linkages, such as gears, shafts, interlocks, and other mechanical drive and control elements.

3.7.3.1 Film loading. The design of the magazine assembly shall facilitate the threading of the film through the magazine when the magazine is detached from the camera body. The design of the magazine shall permit the attachment of new film in the supply cassette to the film remaining in the magazine. Suitable means shall be provided in the magazine assembly to aid in the alignment of the two films and expedite the splicing operation.

3.7.3.2 Magazine counter. The magazine shall contain a seven-digit, additive, and nonresettable counter. The counter shall be located in a visible location on the exterior of the magazine. The counter shall record the total number of camera operation cycles when the magazine is installed on the camera body.

3.7.4 Film cassettes. The camera shall be provided with one supply cassette and one take-up cassette as part of the magazine assembly. The cassettes shall be identical and completely interchangeable with each other. All mechanical and electrical connections shall automatically engage when the cassettes are mated to the camera magazine. The camera magazine shall be provided with permanently legible indicators that indicate the correct position for the supply cassette and the takeup cassette. These indicators shall be visible when the camera is installed in the TARPS. The cassettes shall be light tight to the extent specified in 3.5.3. A data card, 3 inches (76.2mm) wide by 1.5 inches (38.1mm) high, shall be provided on each cassette for making pencil notations. The data card shall be constructed from material that allows the pencil notations to be easily erased.

3.7.4.1 Film capacity. Each cassette shall have a capacity of 2,000 feet (609.6mm) of 5-inch (127mm) film having a nominal base thickness of 2.5 mils and wound on an MS-26565-12 spool.

3.7.4.2 Film supply indicator. Each cassette shall be equipped with a film supply indicator. The indicator shall show the amount of film remaining in the cassette. The indicator shall be in a convenient visible location on the exterior of the cassette.

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3.7.4.3 End-of-film sensor. The magazine shall be provided with an end-of-film sensor. The sensor shall be capable of stopping camera operation at the maximum camera cycle rate when there is not more than 30 feet (9.1m) of film remaining in the supply cassette. The sensor shall provide an end-of-film signal as specified in 3.9.1.2.2

3.7.5 Dust covers. The camera body and magazine assembly shall be provided with easily removable and replaceable dust covers which shall protect mechanisms and optics.

3.8 Environmental conditions. The camera shall operate as required under the conditions specified in 3.8.1 through 3.8.10 (see 4.5.2).

3.8.1 Sand and dust. The contractor shall prepare an environmental report (see 6.2.2), to verify that the camera will withstand sand and dust particle encountered under service conditions in operational areas of the world when installed in the TARPS (see 4.5.2.1).

3.8.2 Fungus. The contractor shall prepare an environmental report (see 6.2.2), to verify that approved non-nutrient materials are used throughout the camera to the greatest extent possible and that all other materials used shall be suitably treated to resist fungus growth as encountered in tropical climates (see 4.5.2.2).

3.8.3 Attitude. The camera shall operate in any attitude or position (see 4.5.2.3).

3.8.4 Temperature-altitude. The camera shall meet the temperature-altitude requirements specified in 3.8.4.1 and 3.8.4.2 (see 4.5.2.4).

3.8.4.1 Equipment operating. The camera shall operate as specified under all combinations of temperature and altitude in the ranges of 0 degree F to 120 degrees F (-17.8 degrees C to 48.9 degrees C), and sea level to 50,000 feet (15,240mm). When operated at 0 degree F (-17.8 degrees C), the camera shall be permitted to use internal heaters for a period of ten minutes prior to operation.

3.8.4.2 Equipment nonoperating. The camera shall withstand all combinations of temperature and altitude in the ranges of -65 degrees F to 160 degrees F (-53.9 degrees C to 71.1 degrees C) and sea level to 50,000 feet (15,240mm).

3.8.5 Humidity. The camera shall withstand the effects of humidities up to 100 percent, including conditions where condensation takes place in and on the camera, except as specified herein (see 4.5.2.5).

3.8.6 Salt atmosphere. The camera shall withstand exposure to salt sea atmosphere under service conditions encountered in sea operations (see 4.5.2.6)

3.8.7 Explosive atmosphere. The camera shall be capable of operating in an explosive atmosphere. The camera shall not cause the ignition of a

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ambient gaseous mixture with air when operated in such an atmosphere (se 4.5.2.7).

3.8.8 Acceleration. The camera shall meet the acceleration requirements specified herein (see 4.5.2.8 and 4.5.2.9).

3.8.8.1 Equipment operating. The camera shall withstand the following steady-state stresses due to acceleration while operating.

Direction of Acceleration Vector	Acceleration G-Level
a. Fore	1.5
b. Aft	1.5
c. Down	6.5
d. Up	11.5
e. Lateral	7.5

3.8.8.2 Equipment nonoperating. The camera shall withstand the following steady-state stresses due to acceleration:

Direction of Acceleration Vector	Acceleration G-Level
a. Fore	10.4
b. Aft	10.4
c. Down	6.5
d. Up	13.8
e. Lateral	7.5

3.8.9 Vibration. The camera shall not be damaged and shall operate as specified when hard mounted and subjected to vibrations as shown in FIGURE 5 and tabulated below (see 4.5.2.10):

20 to 34Hz at .005 g^2/Hz
34 to 46Hz increase at 10.5 db/octave to .015g^2/Hz
46 to 70Hz at .015 g^2/Hz
70 to 100Hz decrease at -8.75 db/octave to .0055g^2/Hz
100 to 360Hz increase at +3.0 db/octave to .019g^2/Hz
360 to 700Hz decrease at -5.3 db/octave to .006g^2/Hz
700 to 850Hz increase at 7.5 db/octave to .010g^2/Hz
850 to 1100Hz decrease at -7.7 db/octave to .005g^2/Hz
1100 to 1500Hz increase at 6.4 db/octave to .010g^2/Hz
1500 to 2000Hz decrease at -14.0 db/octave to .0025g^2/Hz

Overall = 4.04 grms (min.).

3.8.10 Shock. The camera, with vibration isolators in place, shall not be damaged and shall meet the performance requirements specified herein when subjected to impact shock of 15g. The camera mounting base with excursion stops or bumpers in place, individual isolators, other attaching devices, and maximum rated load applied in a normal

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manner shall withstand impact shock of 30g. The camera shall be capable of withstanding the shock environment encountered during servicing operations (see 4.5.2.11).

3.9 Performance characteristics. The camera shall perform as specified in 3.9.1 through 3.9.1.1.7.5.

3.9.1 Operation. The warmup time required for the camera prior to operation under service conditions shall be not greater than 10 minutes. The camera shall require no warmup time prior to operation under room ambient conditions. The camera shall require no special cooling. The camera shall not be required to operate when the temperature of the camera exceeds 150 degrees F (65.6 degrees C) while installed in the TARPS. No special preflight thermal stabilization shall be required for proper operation of the camera. Built-in heaters may be used to provide heating during engine startup and during the mission as required. No more than 10 minutes of pre-heating time shall be required prior to camera operation. The camera, when supplied with electrical power and command signals, shall be capable of operating in an autocycle mode at 0.38 to 8 cps with a 55 percent overlap and in a pulse mode at a fixed scanning rate of not less than 0.50 cps, at a cycling rate determined by an externally provided intervalometer signal that is less than, or equal to, 0.40 pulse per second.

3.9.1.1 Input signals. Input signals shall be provided to the camera as specified in 3.9.1.1.1 through 3.9.1.1.7.5.

3.9.1.1.1 Vg/H signal. A positive DC voltage proportional to Vg/H and appropriately scaled to cover all modes of camera operation shall be provided. The scale factor in the normal mode shall be 50.3 (volts per knot per foot). When operating in either standoff mode, the scaled voltage shall be sin 30 degrees times 50.3 (volts per knot per foot) or 25.15 (volts per knot per foot) where 30 degrees represents the angle from the horizon where the overlap from frame to frame shall be 55 percent. The scale change to standoff shall be made external to the camera. For eight cps, the maximum operational voltage shall be 53.7 volts. A return line shall be provided for the Vg/H voltage. Both the signal line and the return line shall have resistive loads greater than 100,000 ohms.

3.9.1.1.2 Cloud cover exposure signals. Two 28VDC cloud cover exposure signals which provide a +1 and +2 stop exposure increase override shall be provided.

3.9.1.1.3 Vehicle altitude signal. A positive voltage proportional to the vehicle altitude (H) shall be provided. The scale factor shall be 0.5 volts per 1,000 feet (304.8m) Above Ground Level (AGL). The maximum H shall be 70,000 feet (21,336m) (35 volts). The load resistance shall be greater than 100,000 ohms. A return line shall be provided for the H voltage. When the loss of altitude signal voltage occurs, the camera shall automatically self-focus to an altitude of 6,000 feet (1,829m) AGL.

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3.9.1.1.4 Intervalometer pulse train signal. An intervalometer pul. train signal shall be provided. The pulse rate shall be appropriately scaled to cover all modes of camera operation. The pulse source shall provide a pulse rate of 7.5Vg/H pulses per second where Vg/H is scaled as a function of the selected mode as specified in 3.9.1.1.1. The pulse amplitude shall be +28 volts and the pulse width shall be 75 plus or minus 25 milliseconds.

3.9.1.1.5 BIT initiate signal. A BIT initiate signal shall be provided. The signal shall be a momentary switch closure to 28VDC with a minimum pulse width of 10 milliseconds.

3.9.1.1.6 Interface with LED RHA. The Light Emitting Diode Recording Head Assembly (LED RHA) at the camera shall be comprised of a linear, 1 x 60 LED array positioned so that successive firings of the LEDs shall produce a BCD data matrix or an alphanumeric data matrix in accordance with FIGURE 3 and FIGURE 4, respectively. The placement and orientation of data blocks in the camera film format between frames of photographic images shall be in accordance with FIGURE 2. The input signals to the LED RHA are in accordance with FIGURE 6, as the LED drive portion of AN/ASQ-172A Data Display Set - camera interconnection block diagram. One set of output signals from a single electrical connector, 1J4 on the AN/ASQ-172A Data Display Set, is routed directly to the camera LED RHA connector J3, through assigned wires in the interconnecting cable in accordance with FIGURE 7. Connector pins and associated input functions for the J3 connector shall be as specified in 3.9.1.3.

3.9.1.1.6.1. LED RHA drive signals. The AN/ASQ-172A Data Display Set shall provide the required pulses to drive the linear, 1 x 60 LED array. The AN/ASQ-172A Data Display Set shall be capable of firing all 60 LEDs simultaneously. Each data laydown shall consist of a maximum of 60 firings repeated a maximum of 40 times. The data laydown shall be all BCD, all alphanumeric, or alternate as determined by the setting of the appropriate controls of the AN/ASQ-172A Data Display Set. A typical LED firing current pulse is shown in FIGURE 8. Any shape pulse which supplies an amount of energy equivalent to the typical pulse within the same adjustable time period shall be acceptable, provided that the maximum current is not greater than 825ma and provided that the annctation is successfully accomplished. The maximum allowable diode voltage drop shall be 4.0 volts. The LED drive circuits shall be designed to be fail-safe. Regardless of the type of failure, current to any LED shall not remain on for more than 50 µsec. Interface circuitry for LED firing shall be in accordance with FIGURE 9.

3.9.1.1.6.2 BCD form. The BCD form shall be accomplished by firing LED rows 5 to 36. A total of 18 firings shall be employed to produce a data matrix.

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3.9.1.1.6.3 Alphanumeric form. The alphanumeric form shall be accomplished by firing all 60 LED rows. A total of 40 firings shall be employed to produce a data matrix. Each alphanumeric character shall be formed by firing seven contiguous LEDs a maximum of five times. Definition of the character line is shown below. LED rows 40, 48, and 56 shall not be utilized. The AN/ASQ-172A Data Display Set is connected to a total of 60 rows.

	Line	LED Rows
a.	8	1 through 7
b.	7	9 through 15
c.	6	17 through 23
d.	5	25 through 31
e.	4	33 through 39
f.	3	41 through 47
g.	2	49 through 55
h.	1	57 through 63

3.9.1.1.7 Camera command signals. The signals specified herein shall consist of 28VDC closures.

3.9.1.1.7.1 Ready command signals. When the signal is de-energized (OFF), the heater, BIT, and interlock open functions shall be operating. When the signal is energized (ON), all camera circuitry shall become operative, except the scan and film transport system.

3.9.1.1.7.2 Operate command signal. When the signal is energized (ON), the scan and film transport functions shall operate providing the photographic performance of the camera system. The heaters and blowers shall be de-energized (OFF) during camera operation.

3.9.1.1.7.3 Air-to-air command signal. When the signal is energized (ON), the camera shall operate at a constant one cps with the range focus set at infinity.

3.9.1.1.7.4 Port standoff command signal. When the signal is energized (ON), the AEC shall respond to the port photosensor only. The Vg/H signal voltage shall be scaled as specified in 3.9.1.1.1. The camera shall operate at one-half the normal autocycle rate.

3.9.1.1.7.5 Starboard standoff command signal. When the signal is energized (ON), the AEC shall respond to the starboard photosensor only. The Vg/H signal voltage shall be scaled as specified in 3.9.1.1.1. The camera cycle rate shall be as specified in 3.9.1.1.7.4.

3.9.1.2 Camera output signals. The camera shall provide 28VDC electrical output signals as specified in 3.9.1.2.1 through 3.9.1.2.5.

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3.9.1.2.1 Film remaining signal. The camera shall provide a 28VDC pulse for each camera cycle. The pulse shall drive a remote frames remaining counter and shall energize a remote 28VDC 500ma lamp load. The 28VDC pulse duration shall be equal to one half of the camera cycle duration. The minimum pulse duration shall be 62 milliseconds at eight cps.

3.9.1.2.2 Film failure signal. The camera shall contain a film failure switch with circuitry which shall transmit a steady-state signal to a remote fail indicator when a film jam, film breakage, or expenditure of film occurs. The signal shall stop camera operation. The signal shall be removed by placing the ready command signal (see 3.9.1.1.7.1) in the "OFF" position.

3.9.1.2.3 Interlock open signal. The signal shall be a 28VDC switch closure capable of supporting a 500ma lamp load. The signal shall indicate end of fim supply, lack of film supply, takeup cassette or supply cassette not properly latched to the magazine, magazine connectors or electronics unit connectors not mated to the camera body, or the remote cable is not properly connected when the electronics unit is remotely located.

3.9.1.2.4 WRA failure indicator signals. The camera shall provide a 28VDC pulse of 75 milliseconds minimum duration. The pulse shall be capable of supporting a 500ma lamp load for each WRA failure. When failures have occurred, the WRA failure indicator signal output(s) shall occur following the initiation of BIT (see 3.9.1.1.5).

3.9.1.2.5 BIT failure signal. The camera shall provide a remote 28VDC pulse of 75 milliseconds minimum duration capable of supporting a 500ma lamp load when a camera failure occurs following the initiation of BIT (see 3.9.1.1.5). The signal shall activate a magnetic latching "NO-GO" indicator located on the electronics unit of the camera.

3.9.1.2.6 Data signals. The data signals shall be as specified in 3.9.1.2.6.1 through 3.9.1.2.6.4.

3.9.1.2.6.1 Data freeze signal. The data freeze signal shall command the AN/ASQ-172A Data Display Set to stop updating information and begin formatting the required BCD or alphanumeric outputs. The data freeze signal and its relationship to the other timing signals shall be in accordance with FIGURE 10. Interface circuitry shall be in accordance with FIGURE 11.

3.9.1.2.6.2 Data print signal. The data print signal shall command the AN/ASQ-172A Data Display Set to begin data output to the camera. The data print signal and its relationship to the other timing signals shall be in accordance with FIGURE 10. Interface circuitry shall be in accordance with FIGURE 11.

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3.9.1.2.6.3 File spacing signal. The file spacing signal shall define 0.018-inch (0.46mm) of film motion to permit proper spacing of the data block columns. The file spacing signal and its relationship to other timing signals shall be in accordance with FIGURE 10. Interface circuitry shall be in accordance with FIGURE 11.

3.9.1.2.6.4 Film sensitivity control signal. The film sensitivity control signal shall be a mission constant DC voltage determined by the camera S/C switch setting. The DC voltage amplitude shall determine the required current from the AN/ASQ-172A Data Display Set to the LED RHA installed in the camera. The required LED current, as a function of film sensitivity control voltage, shall be in accordance with FIGURE 12. Interface circuitry shall be in accordance with FIGURE 13.

3.9.1.3 Interface connectors. Signal and power interface connector J1 Pin designation(s) and function(s) shall be as specified in TABLE II. The camera interface connector J3 Pin designation(s) and function(s) shall be specified in TABLE III.

3.9.2 Modes of operation. The camera shall be capable of operating in the normal mode, the port standoff mode, the starboard standoff mode, and the air-to-air mode in response to remote commands.

3.9.2.1 Normal mode. When in the normal mode, the camera shall be capable of operating in autocycle or pulse. Autocycle operation shall provide a continuous and uniform cycling rate that is proportional to the vehicle supplied Vg/H signal specified in 3.9.1.1.1. FMC shall be provided. Pulse operation shall provide a single, fixed speed camera cycle for each vehicle supplied intervalometer pulse specified in 3.9.1.1.4. The transition between autocycle and pulse operation, as well as FMC engaged/disengaged, shall be internally controlled. The camera shall respond to the proper H signal while in the normal mode of operation.

3.9.2.2 Standoff mode. When in the port standoff mode and the starboard standoff mode, the camera shall operate as specified in the normal mode, except that the driving signals shall be externally rescaled by a constant factor so as to provide stereo overlap of 55 percent at an oblique depression angle of 30 degrees.

3.9.2.3 Air-to-air mode. When in the air-to-air mode, the camera shall operate in autocycle at a fixed rate of 1 cps. FMC shall be disengaged and the slant range servo shall be driven to the infinity focus position.

3.9.3 Continuous range focus correction. The camera shall be capable of continuously correcting for range focus error over an altitude range of 500 feet (152.4m) to infinity. An externally supplied altitude signal shall be provided to the range servo so that range focus correction may be accomplished.

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TABLE II. J1 connector/pin function designations.

Connector Pin	Function
1,2, and 3	A/C 115VØA
4,5, and 6	A/C 115VØB
7,8, and 9	A/C 115VØC
10 and 11	A/C 28VDC
12 and 13	Power Ground
17	Ready Command
18	Operate Command
19	Port Standoff Mode
20	Starboard Standoff Mode
21	Air-to-Air Mode
22	Intervalometer
23	BIT Initiate
24	AEC +1 Stop
25	AEC +2 Stops
32	BIT Fail Indicator Flag
33	Frames Remaining
34	Film/Fail Indicator
35	Interlock Open Indicator
36	Body Fail Indicator Flag
37	Magazine Fail Indicator Flag
38	Electronics Unit Fail Indicator Flag
47	Vg/H Signal Hi
49	Altitude Signal Hi
50	Altitude Signal Return
48	V/H Signal Return

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TABLE III J3 connector pin/function designations.

Connector Pin	Function	Connector Pin	Function
1	Data Freeze +	43	Rows 25-30 Cathode
2	Data Freeze -	44	LED Row 31 Anode
3	Data Print +	45	LED Row 32 Anode
4	Data Print -	46	LED Row 33 Anode
5	File Spacing +	47	LED Row 34 Anode
6	File Spacing -	48	LED Row 35 Anode
7	Film Sensitivity +	49	LED Row 36 Anode
8	Chassis (Gnd)	50	Rows 31-36 Cathode
9	LED Row 1 Anode	51	LED Row 37 Anode
10	LED Row 2 Anode	52	LED Row 38 Anode
11	LED Row 3 Anode	53	LED Row 39 Anode
12	LED Row 4 Anode	54	LED Row 41 Anode
13	LED Row 5 Anode	55	LED Row 42 Anode
14	LED Row 6 Anode	56	LED Row 43 Anode
15	Rows 1-6 Cathode	57	Rows 37-43 Cathode
16	LED Row 7 Anode	58	LED Row 44 Anode
17	LED Row 8 Anode	59	LED ROW 45 Anode
18	LED Row 9 Anode	60	LED Row 46 Anode
19	LED Row 10 Anode	61	LED Row 47 Anode
20	LED Row 11 Anode	62	LED Row 40 Anode
21	LED Row 12 Anode	63	LED Row 50 Anode
22	Rows 7-12 Cathode	64	Rows 44-50 Cathode
23	LED Row 13 Anode	65	LED Row 51 Anode
24	LED Row 14 Anode	66	LED Row 52 Anode
25	LED Row 15 Anode	67	LED Row 53 Anode
26	LED Row 16 Anode	68	LED Row 54 Anode
27	LED Row 17 Anode	69	LED Row 55 Anode
28	LED Row 18 Anode	70	LED Row 57 Anode
29	Rows 13-18 Cathode	71	Rows 51-57 Cathode
30	LED Row 19 Anode	72	LED Row 58 Anode
31	LED Row 20 Anode	73	LED Row 59 Anode
32	LED Row 21 Anode	74	LED Row 60 Anode
33	LED Row 22 Anode	75	LED Row 61 Anode
34	LED Row 23 Anode	76	LED Row 62 Anode
35	LED Row 24 Anode	77	LED Row 63 Anode
36	Rows 19-24 Cathode	78	Rows 58-63 Cathode
37	LED Row 25 Anode	79	Chassis (Gnd)
38	LED Row 26 Anode		
39	LED Row 27 Anode		
40	LED Row 28 Anode		
41	LED Row 29 Anode		
42	LED Row 30 Anode		

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3.9.4 Relative shutter speeds. The camera relative shutter speed ranges shall be 1/43 to 1/1070 second at 0.38 cps and 1/900 to 1/22,600 second at 8 cps.

3.9.5 Camera system resolution. The camera positioned at infinity focus, loaded with EK3412 (PAN-X) film, a Wratten No. 12, or equal, filter over the lens, and utilizing high contrast (1000:1) resolution targets in accordance with MIL-STD-150, shall provide the minimum average dynamic bench resolution in the image area as specified herein.

3.9.5.1 FMC disengaged. With FMC disengaged, the camera shall provide the following minimum average dynamic bench resolution in the image area:

Mode and cycle rate (cps)	Shutter speed (sec)	Resolution (lp/mm)
Autocycle		
0.50	1/1000	80
3	1/2000	75
8	1/2000	60
Pulse		
0.50	1/1000	75

3.9.5.2 FMC engaged. With FMC engaged, the camera shall provide the following minimum average dynamic bench resolution in the image area:

Mode and cycle rate (cps)	Shutter speed (sec)	Resolution (lp/mm)
Autocycle		
0.50	1/1000	71
3	1/2000	67
8	1/2000	53

3.10 Electrical requirements. The camera shall operate from AC and DC power as specified in 3.10.1 through 3.10.1.2.

3.10.1 Input power. The camera shall meet all applicable requirements of MIL-STD-704 and shall be capable of providing the specified performance from power sources with characteristics as specified in MIL-STD-704. The contractor shall indicate the effects of voltage transients on camera performance in accordance with MIL-STD-704. The input power limitations are as follows:

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a. AC power (three phase) 115/200 volt 400 Hz
Camera system steady state - 1450 watts maximum
Heaters - 400 watts maximum
Unbalance limits for 3 ø power shall be as specified in MIL-STD-704.

b. DC power
Operating voltage limits - 24 to 28.5 volts, category B, steady state - 50 watts maximum.

3.10.1.1 Undervoltage protection. The camera shall not be damaged when subjected to the minimum voltages specified in TABLES I and II of MIL-STD-704. The camera shall automatically resume normal operation when the voltage returns to within normal limits.

3.10.1.2 Electrical overload protection. Each internal regulated power supply in the camera shall be of the current-limiting type, and shall be capable of continuous operation at any load condition between no load and short circuit conditions. The voltage of a regulated supply shall not decrease for currents up to 130 percent of the maximum current used from the supply. When an overload condition ceases to exist for a regulated power supply, the supply shall automatically resume normal regulation within 0.5 second. Each internal unregulated power supply shall be protected against damage from current overload conditions.

3.10.2 Electrical interface connectors. Electrical interface connections between the camera body and magazine assembly, and the camera body and electronics unit shall be made through connectors in the mating surfaces that shall make proper electrical contact when the mating surfaces are engaged. The electrical interface connectors shall meet the requirements of MIL-C-24308 and shall be in accordance with FIGURE 14. The design of the connectors shall be such that reverse and improper assembly, mating, and installation is not possible.

3.10.3 Cabling and connections. Cabling and connections shall be as specified in 3.10.3.1 through 3.10.3.3.

3.10.3.1 Interconnecting cabling. The camera shall meet the requirements specified herein when operated with external wiring in accordance with the applicable requirements of MIL-W-5088. Airplane cabling between the camera and other airplane systems will be provided by the procuring activity. The external wiring shall be unshielded and an overshield may be used when required to meet interference control requirements, and provided the assembly of the cable to its plugs may be easily accomplished. Interconnecting signals shall be differential and shall not depend on mounting structure for signal transmission.

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3.10.3.2 Wire connectors. Wire connectors shall be utilized to the greatest extent possible to preclude the cutting or unsoldering of wires when dismantling the camera, replacing major subassemblies, and reassembling the camera. Connector shall be keyed to prevent the improper insertion of a connector or subassembly into the wrong location within or on the camera.

3.10.3.3 External electrical connectors. External electrical connectors shall be selected in accordance with MIL-STD-1353 and shall be in accordance with FIGURE 14.

3.10.4 Circuit identification. Circuits shall be identified in accordance with MIL-STD-681. Circuit tracing throughout the camera shall be accomplished without the use of the auxiliary means.

3.11 Electromagnetic interference. The camera shall meet the electromagnetic interference requirements of MIL-STD-461 for Class IC equipment.

3.12 Dimensions. The maximum overall dimensions of the camera, excluding the cradle assembly specified in 3.3, shall be not greater than 36.20 inches (919.5mm) in length, 20.68 inches (525.3mm) in width and 21.20 inches (538.48mm) in height).

3.13 Weight. The weight of the camera loaded with 2,000 feet (609.6m) of 5-inch (127mm) film, excluding cables and the cradle assembly specified in 3.3, shall be not greater than 240 pounds (108.9kg).

3.14 Finish. The exterior and interior finishes of the camera shall be as specified in 3.14.1 and 3.14.2.

3.14.1 Exterior. All exterior surfaces of the camera, except as specified herein, shall have a corrosion and abrasion resistant finish in accordance with MIL-E-5558, TYPE I, wrinkle-tinted. The color shall be in accordance with FED-STD-595, Number 16473. Machine surfaces used for the purpose of accurately mounting parts shall be left in their smooth state. Machine surfaces shall be protected by an electrolytic or chemical finish when feasible. Except when prevention of radio interference is required, the machine surfaces shall be protected by a light application of high grade, hard industrial wax.

3.14.2 Interior. All interior surfaces of the camera shall be given a nonreflecting, dull, and durable black finish wherever practicable. The provisions of 3.14.1 pertaining to machine surfaces shall apply.

3.15 Identification of product. The camera shall be marked for identification in accordance with MIL-STD-130.

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3.15.1 Nomenclature and nameplates. Nomenclature assignments and nameplate approval for camera identification shall be in accordance with MIL-N-18307.

3.15.2 Serial number. A serial number shall be assigned to each major component. The serial number shall not be duplicated for any similar piece of equipment.

3.15.3 Stock number. Upon request of the contractor, the stock number shall be supplied by the contracting officer.

3.16 Drawings. When specified in the contract (see 6.2.1), a complete set of Level 3 drawings and all associated lists as specified in DOD-D-1000 shall be prepared (see 6.2.2).

3.16.1 Microfilm. When specified in the contract (see 6.2.1), microfilm records of the drawings and associated data lists shall be prepared as specified in MIL-M-9868 (see 6.2.2).

3.17 Container. A durable container shall be provided to protect the camera and accessories during shipment and storage.

3.18 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

3.19 Government furnished property. The government shall provide one cradle assembly, Naval Avionics Center Part Number 1213AS140, for each KA-99A camera. The cradle is to be assembled to the camera prior to delivery of the equipment to the government.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classifed as follows:

1. First Article Inspection (see 4.5).
2. Quality Conformance Inspection (see 4.6).

4.3 Inspection conditions. Unless otherwise specified in the contract (see 6.2.1) all inspections shall be performed in accordance with the inspection conditions specified in 4.3.1 and 4.3.2.

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4.3.1 Temperature and humidity. Unless otherwise specified in the contract (see 6.2.1) inspection shall be conducted in an ambient ambient temperature of 77 degrees F (25 degrees C), plus or minus 5 degrees, with any relative humidity up to 90 percent.

4.3.2 Input power. The inspections shall be conducted with nominal 28VDC and nominal 115VAC, 400Hz, single phase power sources.

4.4 Failure during inspection. During any of the inspections specified herein, the camera shall not exhibit characteristics which would cause the camera to fail to meet the requirements of Section 3. Characteristics to be considered are deterioration, power loss, sticking, excessive power drain, corrosion, etc.

4.4.1 Corrective validation. When a failure occurs, the individual inspection requirements schedule shall include an inspection to check all cameras for the noted nonconformance until determination has been made that the defect has been corrected.

4.5 First article inspection. The first article inspection shall be as specified in 4.5.1.

4.5.1 First article samples. The first article samples shall consist of three models representative of the production camera. Unless otherwise specified in the contract (see 6.2.1), the schedule for the first article inspection shall be in accordance with TABLE IV.

TABLE IV. First Article Inspection

Inspection	Camera No. 1	Camera No. 2	Camera No. 3
Sand and Dust		X	
Fungus		X	
Attitude			X
Temperature-Altitude		X	
Humidity		X	
Explosive Atmosphere	X		
Acceleration (equipment operating)	X		
Acceleration (equipment nonoperating)	X		
Vibration	X		
Shock	X		
Electromagnetic Interference		X	
Salt-Fog		X	
Reliability Demonstration			X
Maintainability Demonstration			X

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4.5.2 Environmental inspections.

4.5.2.1 Sand and dust. When specified in the contract (see 6.2.1), the camera shall be inspected in accordance with Method 510.1, Procedure I of MIL-STD-810.

4.5.2.2 Fungus. When specified in the contract (see 6.2.1), the camera shall be inspected in accordance with Method 508.1, Procedure I of MIL-STD-810.

4.5.2.3 Attitude. The camera, loaded with 2,000 feet (609.6m) of 5-inch (127mm) film, shall be operated at a cycling rate of one cps for a period of two minutes in each of six positions. After operation, the camera and film shall be inspected to determine that the camera and film have sustained no damage as a result of operating in each of the six positions.

4.5.2.4 Temperature-altitude. The camera shall be inspected in accordance with Method 504.1 of MIL-STD-810 for equipment category 5, except as modified herein. The steps of Table 504.1-II shall be amended as follows:

Step 1b - The chamber temperature shall be adjusted to -65 degrees F (-53.9 degrees C).
Step 2 - The chamber temperature shall be adjusted to zero degree F (-17.8 degrees C).
Step 3 - The chamber temperature shall be adjusted to zero degree F (-17.8 degrees C).
Step 6 - The chamber temperature shall be adjusted to 160 degrees F (71.1 degrees C).
Step 8 - The chamber temperature shall be adjusted to 150 degrees F (65.6 degrees C).

4.5.2.5 Humidity. The camera shall be inspected in accordance with Method 507.1 Procedure I of MIL-STD-810. At the conclusion of the 10th cycle of Step 6 and within one hour after removal of the camera from the chamber, the camera shall be operated to determine that the camera performs as specified in Section 3.

4.5.2.6 Salt-fog. The camera shall be inspected in accordance with Method 509.1, Procedure I of MIL-STD-810, except that the chamber temperature shall be maintained at 75 degrees F (23.9 degrees C) and the inspection duration shall be 24 hours.

4.5.2.7 Explosive atmosphere. The camera shall be inspected in accordance with Method 511.1, Procedure I of MIL-STD-810 using 100/130 octane gasoline. Step 1 shall be performed at 120 degrees F (48.9 degrees C).

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4.5.2.8 Acceleration (equipment operating). The camera shall be inspected in accordance with Method 513.2, Procedure II of MIL-STD-810. The following levels shall be applied during first article testing:

Direction of Acceleration Vector	Acceleration G-Level
a. Fore	1.5
b. Aft	1.5
c. Down	6.5
d. Up	11.5
e. Lateral	7.5

The vehicle category for TABLE 513.2-II shall be aircraft carried stores.

4.5.2.9 Acceleration (equipment nonoperating). The camera shall be inspected in accordance with Method 513.2, Procedure I of MIL-STD-810. The following levels shall be applied during first article testing:

Direction of Acceleration Vector	Acceleration G-Level
a. Fore	10.4
b. Aft	10.4
c. Down	6.5
d. Up	13.8
e. Lateral	7.5

The vehicle category for TABLE 513.2-I shall be aircraft carried stores.

4.5.2.10 Vibration. The camera shall be inspected in accordance with Method 514.2, Procedure IIA of MIL-STD-810, except that the test levels for the camera shall be as shown in FIGURE 15 and tabluated below. The inspection duration shall be one hour functional and one hour endurance. The inspection sequence shall be the functional inspection performed for 0.5 hour, followed by one hour of the endurance inspection, followed by the final 0.5 hour of the functional inspection.

Endurance Level

20 to 50Hz at .05g^2/Hz
50 to 100Hz decrease at 10db/octave to .005g^2/Hz
100 to 500Hz at .005g^2/Hz
500 to 1500Hz increase at 3.8db/octave to .02g^2/Hz
1500 to 2000Hz decrease at -4.25db/octave to .013g^2/Hz
Overall = 4.98 grms (min.)

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Functional Level

20 to 50Hz at .015g /Hz
50 to 100Hz decrease at 10db/octave to .0015g /Hz
100 to 500Hz at .0015g /Hz
500 to 1500Hz increase at 3.4db/octave to .005g /Hz
1500 to 2000Hz decrease at -4.2db/octave to .0034g /Hz
Overall = 2.54 grms (min.)

4.5.2.11 Shock. The camera shall be inspected in accordance with Method 516.2, Procedures I, III, and V of MIL-STD-810. Procedure V shall apply only to the camera electronics unit. The shock pulse shape for Procedures I and III shall be in accordance with Figure 516.2-2 of MIL-STD-810.

4.5.2.12 Electromagnetic Interference. The camera shall be inspected in accordance with the test methods of MIL-STD-462 specified for Class IC equipment of MIL-STD-461. The test method RS03 frequency range shall be 14 kilohertz (KHz) to 12.4 gigahertz (GHz) at 200 volts/meter. Test Methods RS01, RE01, AND RE04 are not required. Broadband CE01 and CE02 tests shall be conducted for engineering data only. Narrowband limits of MIL-STD-461. Class A equipment shall apply to Test Methods CE01, CE02, CE03, CE04, and RE02.

4.5.3 Reliability demonstration. The camera shall be inspected in accordance with Test Plan XIIIC of MIL-STD-781 modified as follows:

Temperature Range:	-20 degrees F to 120 degrees F (-28.9 degrees C to 48.9 degrees C)
Random Vibration:	Acceleration Power Spectral Density = .012g /Hz Frequency = 20 - 2000Hz
Environmental Test Cycle:	IAW FIGURE 16
Camera Operation:	IAW FIGURE 16
Test Duration:	30,000 Camera Cycles (Fixed length, Test Plan XIIIC)
Length of Test:	Approximately 48 days. Also, Test Plan XIIIC of MIL-STD-781 shall be modified to convert hours to cycles in accordance with 3.4.

4.5.4 Maintainability demonstration. The camera shall be inspected for schedule maintenance and unscheduled maintenance at the organizational level and unscheduled maintenance at the intermediate

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level. The inspection shall be performed in accordance with Method 9 of MIL-STD-471 of the 95th percentile level for scheduled maintenance at the organizational level and Method 8 of MIL-STD-471 for unscheduled maintenance at both the organizational and intermediate levels.

4.6 Quality conformance inspection. Each camera submitted for acceptance on the contract or order shall have passed the quality conformance inspections specified in 4.6.1 through 4.6.18.

4.6.1 Physical examination. Examine each camera visually and with gauges to determine conformance to specified requirements for materials, construction, fit, and interchangeability of components, dimensions, weight, finish, markings, wiring, color coding and other visible defects. Operate all controls, releases, locks, and moving parts to determine suitability for the intended purpose. Examine each camera for ease of attachment and detachment of unit parts.

4.6.2 Operational. Each camera shall be inspected in accordance with procedures prepared by the contractor and acceptable to the procuring activity (see 6.2.2), to determine that the camera meets the specified requirements of 3.3, 3.7, 3.9, and 3.10 for BIT, film format, optical barrel, scan and film transport drive servo, film transport mechanism, camera cycle counter, electronics, and electro-mechanical devices, camera electronics unit, magazine counter, film footage indicator, end-of-film sensor, warm-up time, cycling rates, input signals, camera output signals, modes of operation, and input power during live film operation at various camera cycle rates.

4.6.3 Camera light-tightness. Each camera, assembled as a complete unit and loaded with unexposed film having an AFS of 200, shall be exposed to the illumination of a 17,800 Lumen No. 2 photoflood lamp, or equal, on each of the six sides of the camera. The photoflood lamp shall be placed at a distance of five feet from the camera. The total time of exposure to the illumination of the photoflood lamp shall be thirty minutes divided equally between the six sides of the camera. Upon completion of the procedure, the film shall be processed in accordance with the film manufacturer's recommendations. The film shall be inspected for evidence of a light leak in the camera. Evidence of a light leak shall be cause for rejection.

4.6.4 Cassette light-tightness. Each cassette, loaded with unexposed film having an AFS of 200, shall be exposed to the illumination of a 17,800 Lumen No. 2 photoflood lamp on each of the cassettes' four sides. The four sides of the cassettes shall be the two ends, and two nonoverlapping 180-degree sections of the circumference of the cassette. The photoflood lamp shall be placed at a distance of five feet from the cassette. The total time of exposure to the illumination of the photoflood lamp shall be 20 minutes divided equally between the four sides of the cassette. During exposure to

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one section of the circumference of the cassette, the illumination of the photoflood lamp shall be directed at the cassette light trap. Upon completion of the procedure, the film shall be processed in accordance with the film manufacturer's recommendations. The film shall be inspected for evidence of fog exposure. Evidence of fog exposure beyond the first four wraps of film shall be cause for rejection.

4.6.5 Forward motion compensation. Each camera shall be inspected to determine conformance to the FMC requirements of 3.7.1.4, and 3.9.2.1. The inspection shall be performed in accordance with procedures prepared by the contractor and acceptable to the procuring activity (see 6.2.2), except as specified herein. A resolution inspection with FMC engaged shall be performed. The camera shall be loaded with unexposed film having an AFS of 40. The FMC mechanism shall be engaged and the camera shall be operated at cycling rates of 0.50, 3, and 8 cps. The radial and tangential resolution values shall be combined in accordance with the expression, $\sqrt{RT}$, where R = line pairs/mm for the radial portion of the resolution target, and T = line pairs/mm for the tangential portion of the resolution target. The average degradation in resolution shall be not greater than one pattern over the non-FMC resolution inspection result.

4.6.6 Automatic exposure control. Each camera shall be inspected to determine conformance to the requirements of 3.7.1.5 for the shutter and lens diaphragm to provide the proper exposure by responding to AEC functions for controlled inputs of brightness, cycling rate, and S/C settings. The inspection shall be performed in accordance with procedures prepared by the contractor and acceptable to the procuring activity (see 6.2.2), except as specified herein. Suitable calibrated light sources shall be utilized in the inspection procedure. The AEC shall be inspected in three modes of operation--normal, starboard standoff, and port standoff. When in the standoff mode of operation, only the photocell viewing the standoff position selected shall be energized. When in the normal mode of operation, both the port and starboard photocells shall be energized to average scene brightness. When measuring shutter slit response, each of four brightness levels (50, 500, 2500, and 5000 foot-lamberts) shall be applied to the photocell(s) and the response of the shutter slit measured and recorded. The camera shall be operated at each of four cycle rates (0.38, 1, 4, and 8 cps) and the response of the shutter slit shall be measured and recorded. When measuring the response of the lens diaphragm for a given scene brightness level, the S/C switch shall be set at each of four positions (50, 100, 200, and 600) and the response of the diaphragm measured and recorded.

4.6.7 Camera data recording. Each camera shall be inspected to determine conformance to the requirements of 3.7.1.6. The camera shall be loaded with unexposed EK3412 film. The camera shall be operated in conjunction with the AN/ASQ-172A Data Display Set or other

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equipment that is capable of simulating the specified data inputs to the camera. The exposed film shall be processed in accordance with the film manufacturer's recommendations. The film shall be inspected to determine that the format, dimensions, and locations of the alphanumeric data block, BCD data block, numeric frame count, center of format fiducial marks, and direction of flight/direction of film travel marks are in accordance with FIGURE 2.

4.6.8 Lens. Each lens shall be inspected to determine conformance to the requirements of 3.7.1.9 and 3.7.1.9.2. The inspection shall be performed in accordance with the applicable methods of MIL-STD-150 and procedures prepared by the contractor and acceptable to the procuring activity (see 6.2.2).

4.6.9 Antireflection coating. Each lens shall be inspected in accordance with the procedures of MIL-C-675.

4.6.10 Filters. Each filter shall be inspected in accordance with the procedures of MIL-STD-1278.

4.6.11 Mirrors. Each mirror shall be inspected for spectral reflection characteristics and figures in accordance with the procedures of MIL-M-13508.

4.6.12 Optical System. Each complete optical system shall be inspected to determine conformance to the requirements of 3.7.1.11 and 3.7.1.12. The inspected shall be performed in accordance with procedures prepared by the contractor and acceptable to the procuring activity (see 6.2.2).

4.6.13 Continuous Range Focus Correction. Each camera shall be inspected to determine conformance to the requirements of 3.9.3. The inspection shall be performed in accordance with procedures prepared by the contractor and acceptable to the procuring activity (see 6.2.2).

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4.6.14 Relative shutter speeds. Each camera shall be inspected to determine conformance to the requirements of 3.9.4. The inspection shall be performed in accordance with procedures prepared by the contractor and acceptable to the procuring activity (see 6.2.2).

4.6.15 System resolution. Each camera shall be inspected to determine conformance to the requirements of 3.9.5. The inspection shall be performed in accordance with procedures prepared by the contractor and acceptable to the procuring activity (see 6.2.2)

4.6.16 Reliability assurance. The reliability acceptance test shall be performed as follows:

a. A random vibration screen shall be applied to each production unit. The units shall be hard-mounted to a shake table and shaken sequentially in each of three orthogonal axes for a least ten minutes to power spectral density constant at .0126g^2/Hz from 20 to 2000Hz, resulting in a reduction of spectral level to 5.0G.

b. The random vibration screen shall be followed by ten temperture cycles with the last two temperture cycles failure-free. The temperature cycles and equipment ON and OFF time shall be as defined in FIGURE 17.

c. All failures occurring during the vibration and temperature screen shall be evaluated and corrective actions determined and implemented.

4.6.17 Undervoltage protection. Each camera shall be inspected to determine conformance to the requirements of 3.10.1.1. The inspection shall be performed in accordance with procedures prepared by the contractor and acceptable to the procuring activity (see 6.2.2).

4.6.18 Electrical overload protection. Each camera shall be inspected to determine that the internal regulated power supply(s) will handle all load conditions between no load and a short circuit by imposing a short circuit on the power supply(s). Measurements shall be taken and recorded to determine that the regulated power supply(s) is capable of resuming normal regulated operation within 0.5 second after removal of the short circuit. Failure of the regulated power supply to do so shall be cause for rejection. Additional voltage measurements shall be taken to assure that no decrease in the regulated power supply(s) voltage occurs for current values of 130 percent of the required maximum operating current.

4.7 Packaging Inspection. Packaging and packing of the camera shall be examined in accordance with the Quality Assurance Provisions of the documents referenced in Section 5.

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5. PACKAGING

5.1 Preservation and Packaging. Preservation and packaging shall be Level A or Commercial as specified in the contract (see 6.2.1).

5.1.1 Level A. The camera shall be preserved and packaged in accordance with Method IIB of MIL-P-116. Selection and application of preservative shall be at the option of the contractor in accordance with the applicable provisions of MIL-P-116.

5.1.2 Commercial. The camera shall be packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be Level A or Commercial, as specified in the contract (see 6.2.1).

5.2.1 Level A. The camera, packaged as specified in 5.1.1, shall be packed in a Class 2, Style 2, wooden box in accordance with PPP-B-621.

5.2.2 Commercial. The camera shall be packed in a shipping container in a manner that will afford adequate protection against damage during shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract, each unit package, and intermediate and exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use. The camera covered by this specification is intended for use in low and medium altitude reconnaissance photography.

6.2 Ordering Data

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, Number, and Date of this Specification.
b. Number of Cameras Required.
c. First Article Samples Required (see 3.1).
d. Designated Aircraft for Exposure Override (see 3.7.1.5.2)
e. Drawings Required (see 3.17).

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f. Microfilm Required (see 3.17.1).
g. Inspection Conditions if other than specified in 4.3.
h. Temperture and Humidity if other than specified in 4.3.1.
i. First Article Inspection Schedule if other than specified in 4.5.1.
j. Sand and Dust Inspection if Required (see 4.5.2.1).
k. Fungus Inspection if Required (see 4.5.2.2).
l. Level of Packaging and Packing Required (see Section 5).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirements title	Applicable DID No
3.2.3	Report, Reliability Stress Analysis	UDI-R-21423
3.2.4	Contractor Standardization Program Plan	DI-E-7098
3.2.5	Nonstandard Part Approval Requests/ Proposed Additions to An Approved PPSL	DI-E-7028A
3.3.4	Safety Assessment Report System Safety Program Plan Hazard Analysis Report	DI-H-7049 DI-H-7047 DI-H-7048
3.4.1.1	Reliability Program Plan (Production)	DI-R-7079
3.4.1.2	Report, Reliability Prediction	DI-R-7082
3.6.5.2	Report, Maintainability Prediction	DI-R-7108
3.4.1.3	Report, Analysis, Failure Modes and Effects	DI-R-7085
3.4.1.4	Report, Failure Summary and Analysis	DI-R-7041
3.6.4.5	Report, Level of Repair Summary Plan, Level of Repair Program Report, Level of Repair Analysis	DI-L-2082A DI-R-2084A DI-L-2085A

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3.6.5.1	Plan, Maintainability Program	DI-R-7103
3.8.1/3.8.2	Report, Environment, Reliability	DI-R-2116
3.17/3.17.1	Drawings, Engineering and Associated Lists	DI-E-7031
4.5.3	Reports, Reliability Test and Demonstration	DI-R-7034
4.6.2/4.6.5/	Test Procedures	UDI-T-21347
4.6.6/4.6.8/ 4.6.12/4.6.13/ 4.6.14/4.6.15/ 4.6.16/4.6.17	Procedures, Reliability Tests	DI-R-7035

(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L., Vol, II, AMSDL. Copies of data item descriptions required by the contrctors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer).

6.3 First article. When a first article is required for inspection and approval (see 3.1), the first article sample shall consist of three cameras (see 4.5.1).

6.4 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence should apply:

(a) Contract. The contract should have precedence over any specification.

(b) This Specification. This specification should have precedence over all applicable subsidiary specifications.

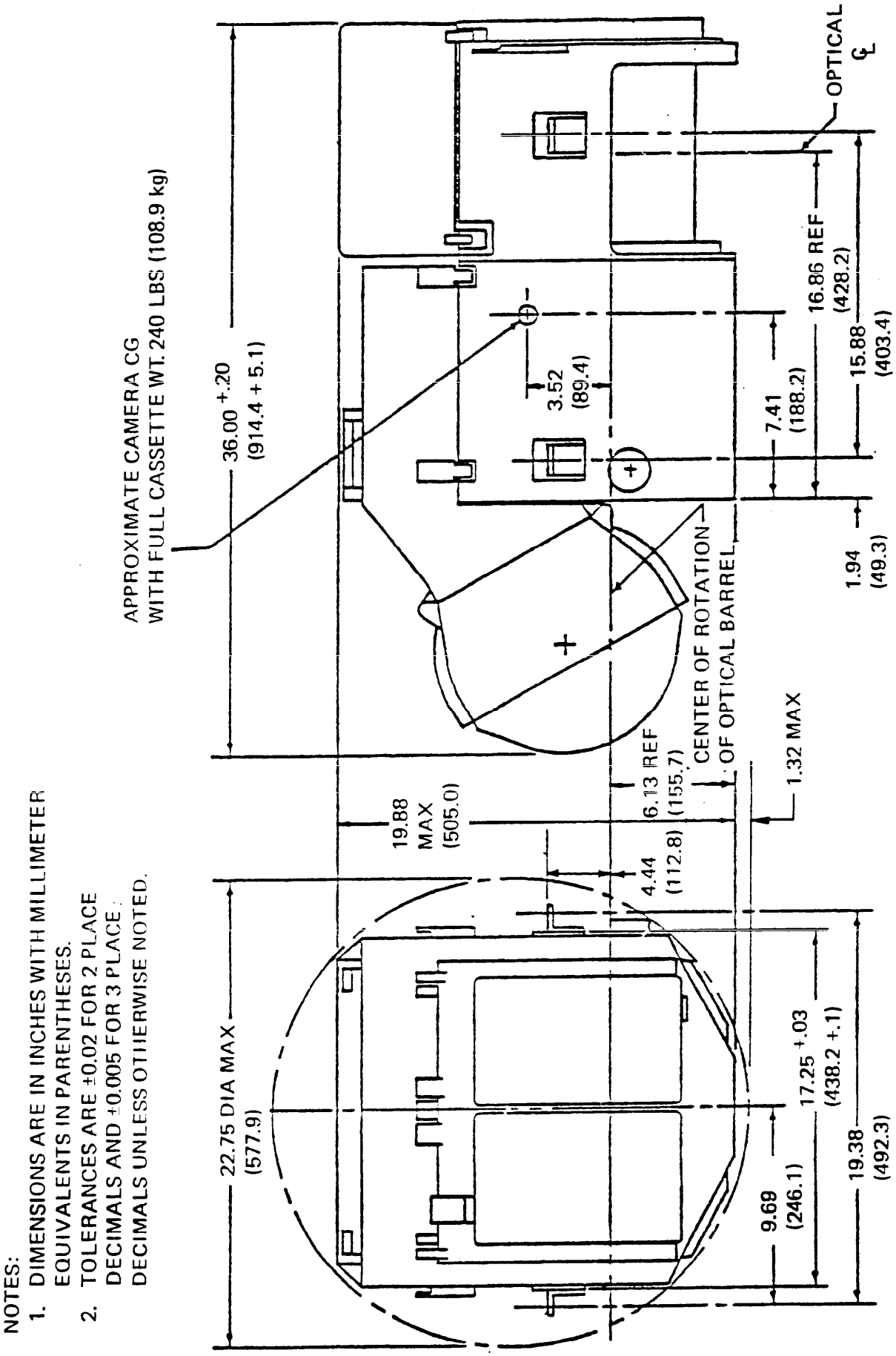
(c) Referenced Specifications. Any referenced specification should have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications should apply to the extent specified.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Government furnished property. The contracting officer should arrange to furnish the property listed in 3.19.

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NOTES:
1. DIMENSIONS ARE IN INCHES WITH MILLIMETER EQUIVALENTS IN PARENTHESES.
2. TOLERANCES ARE ±0.02 FOR 2 PLACE DECIMALS AND ±0.005 FOR 3 PLACE DECIMALS UNLESS OTHERWISE NOTED.

FIGURE 1. KA-99A panoramic camera outline.

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NOTES:
1. DIMENSIONS ARE IN INCHES WITH MILLIMETER EQUIVALENTS IN PARENTHESES.
2. TOLERANCES ARE ±0.04 FOR 2 PLACE DECIMALS AND ±0.005 FOR 3 PLACE DECIMALS UNLESS OTHERWISE NOTED.
3. DEIMENSIONS REFERRING TO DATA BETWEEN FORMATS ARE BASED UPON DOT CENTERS AND/OR ROW LINES.

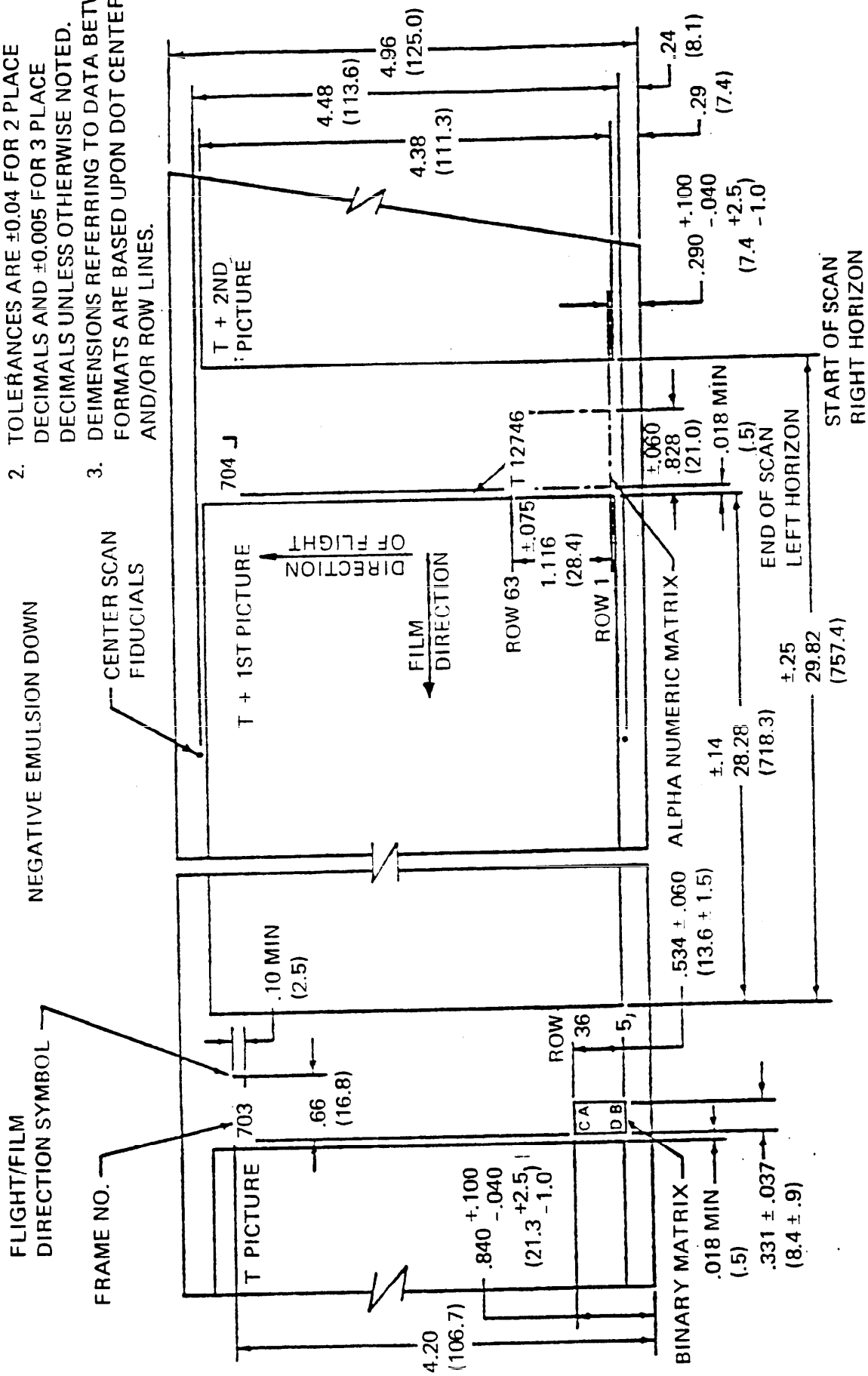


FIGURE 2. Ilm format.

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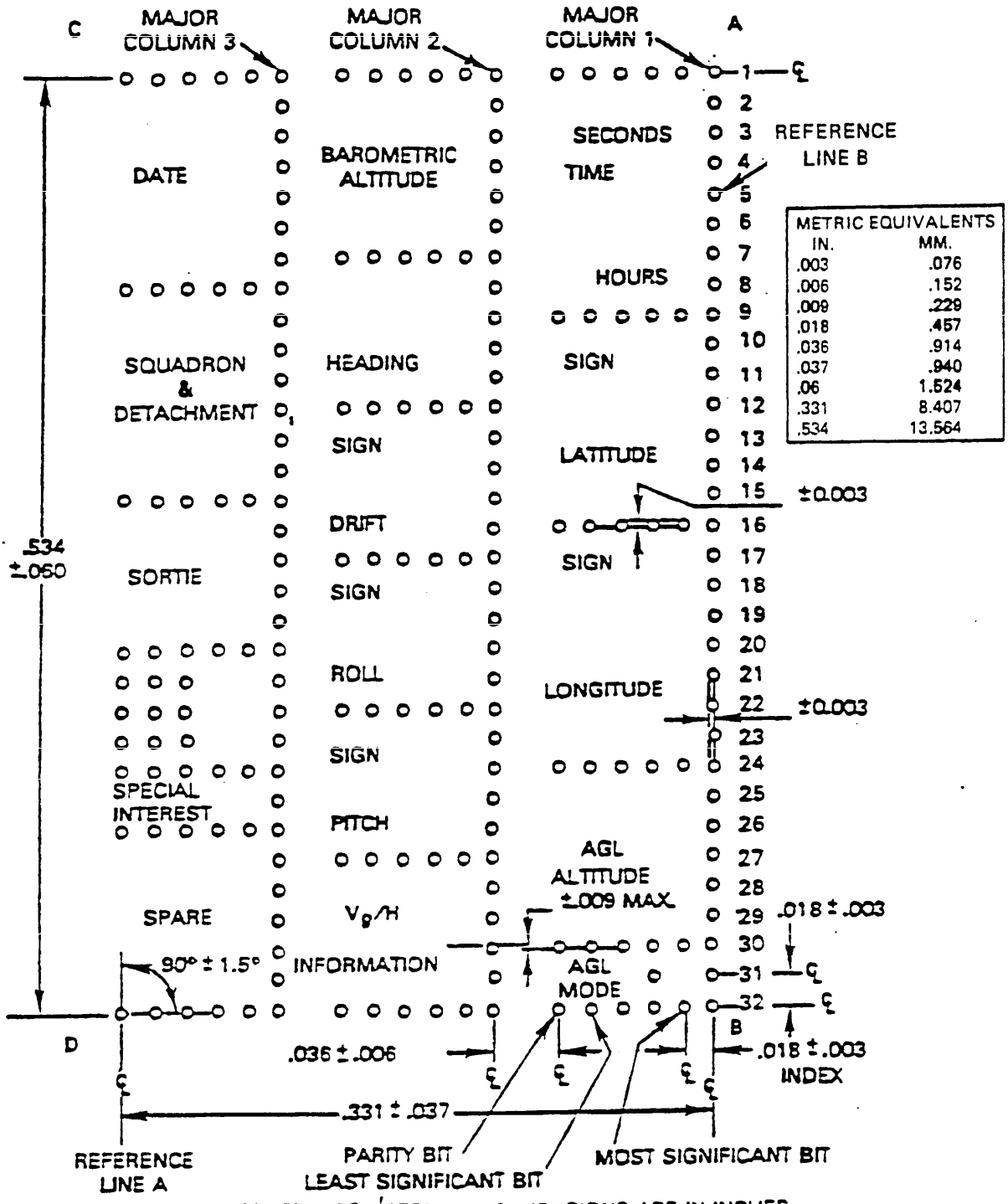


FIGURE 3. BCD data matrix format.

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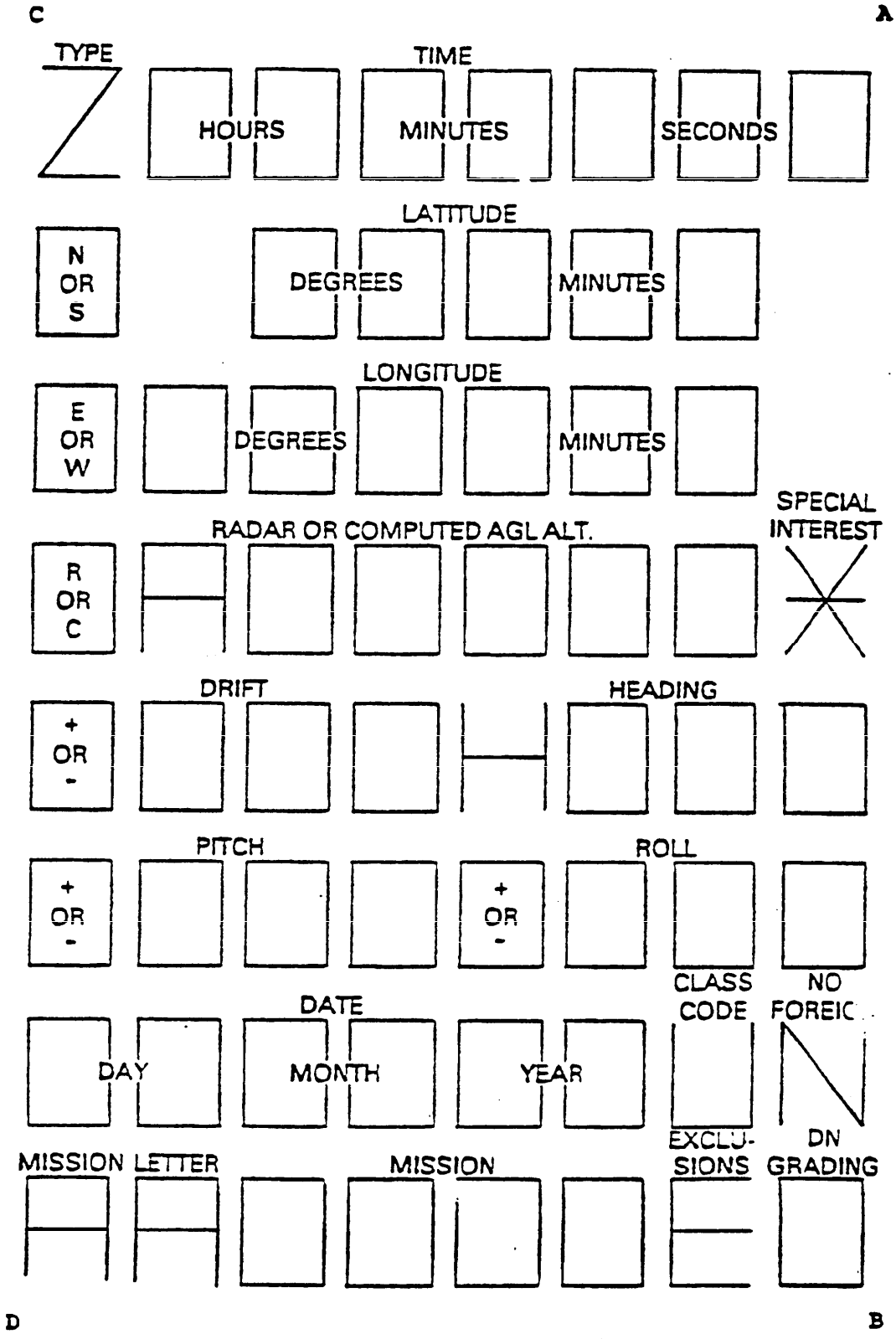
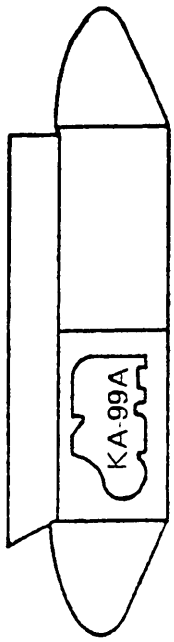


FIGURE 4. A-N data matrix format.

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NOTE: THIS CURVE DERIVED FROM FLIGHT TEST DATA; THEREFORE, TOLERANCE ON THIS CURVE -0 dB +VENDER'S SYSTEM CAPABILITY GRMS = 4.04 (MIN)

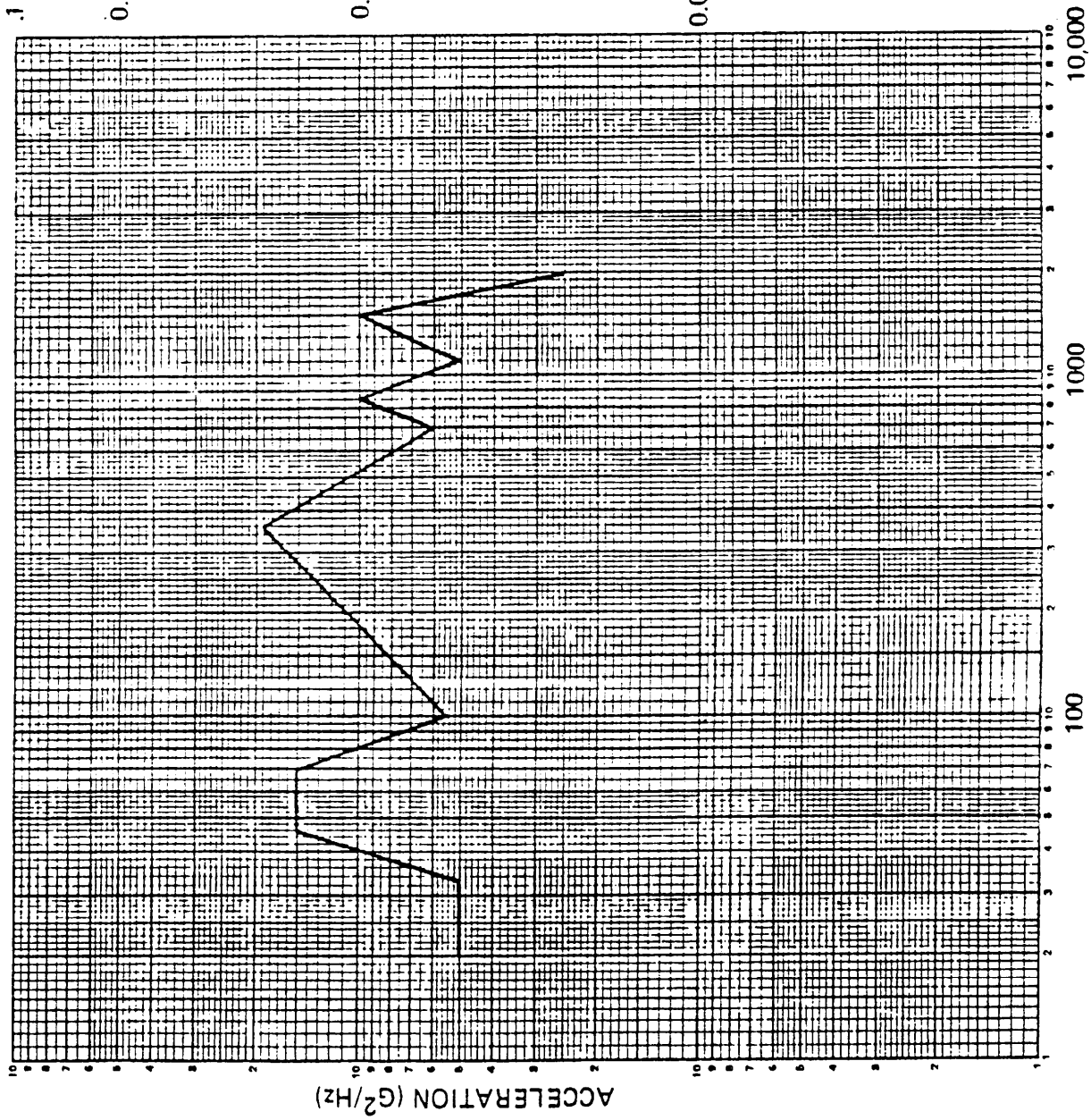


FIGURE 5. Vibration design curve.

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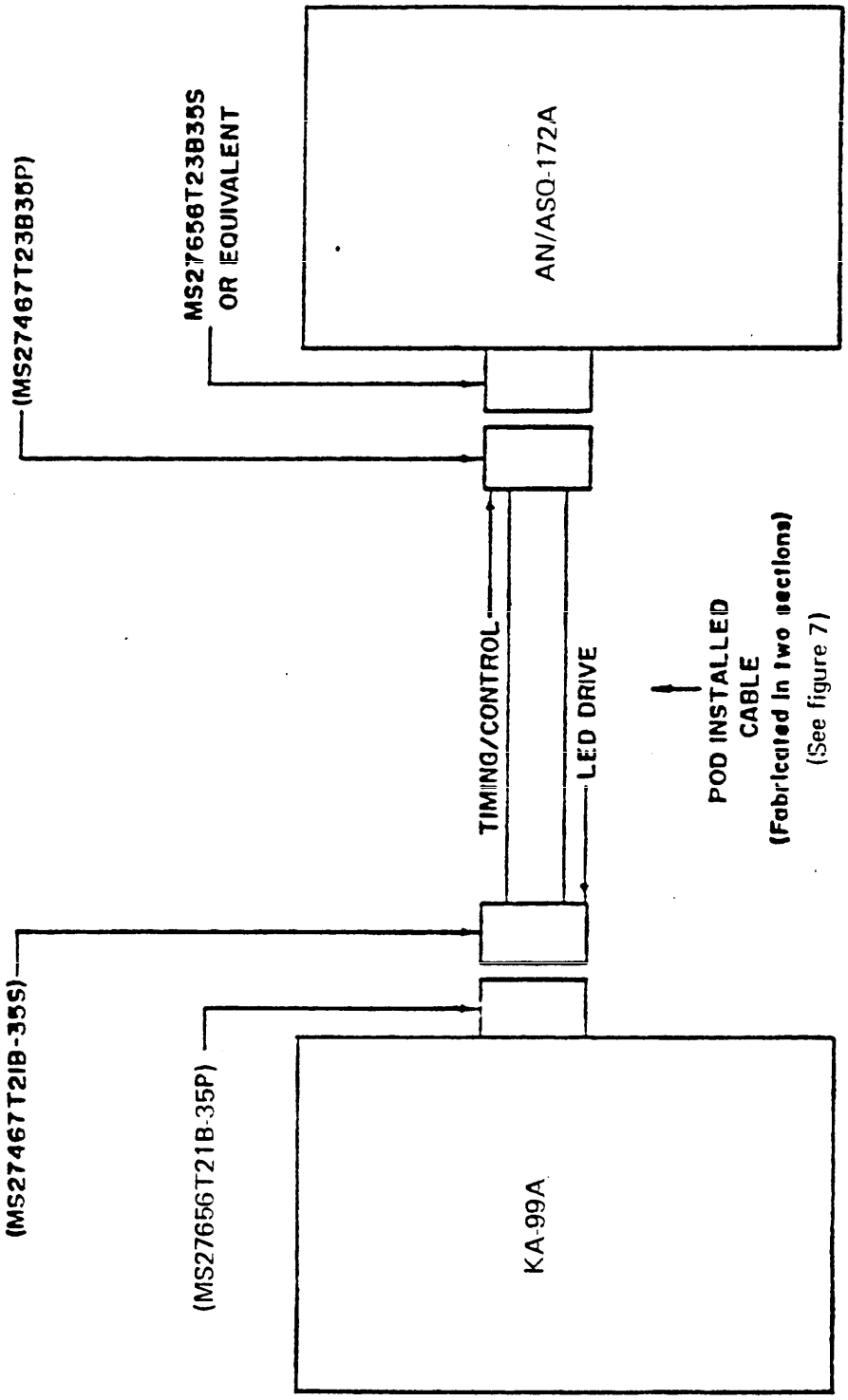


FIGURE 6. KA-99A - AN/ASQ-172A interconnection block diagram.

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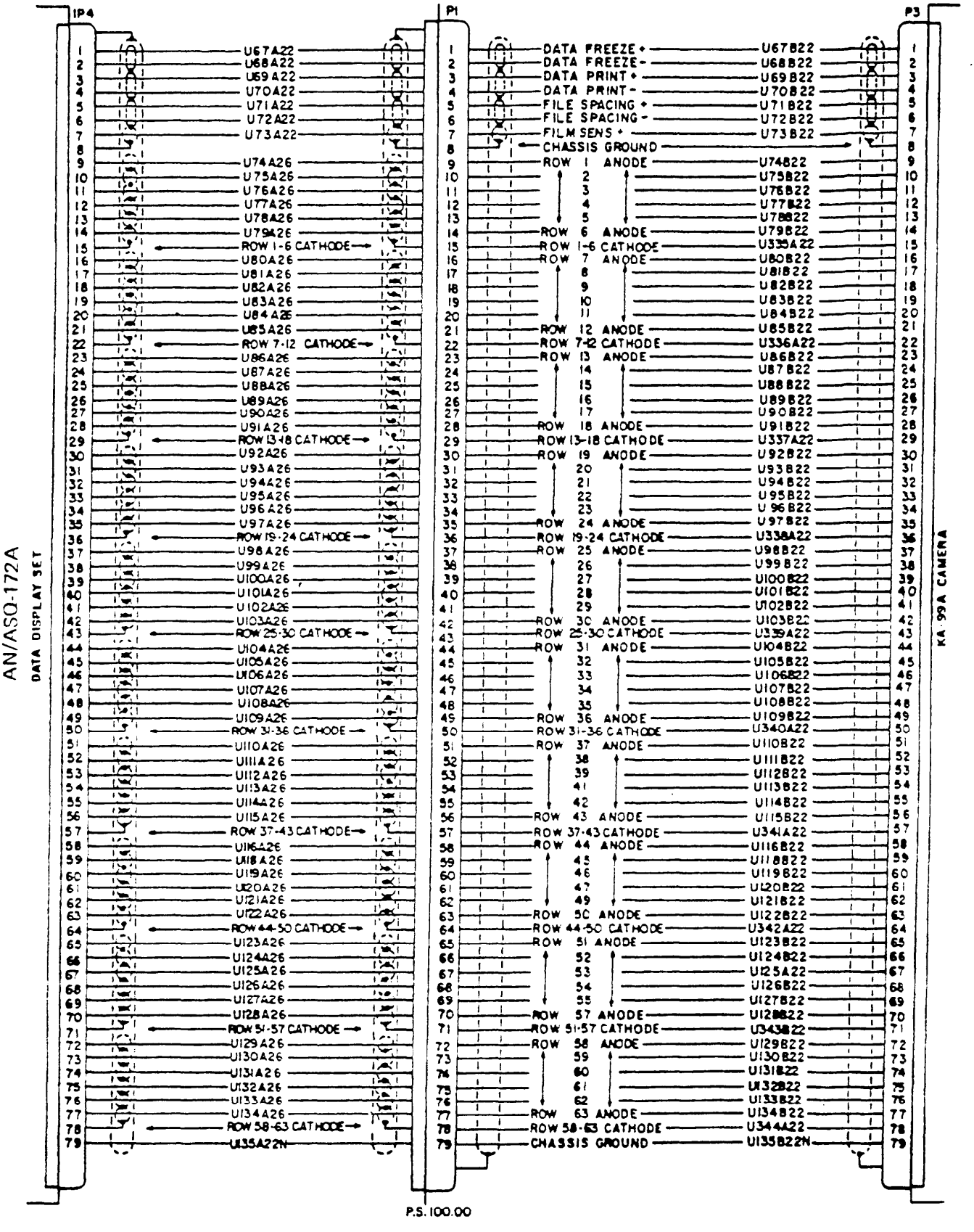
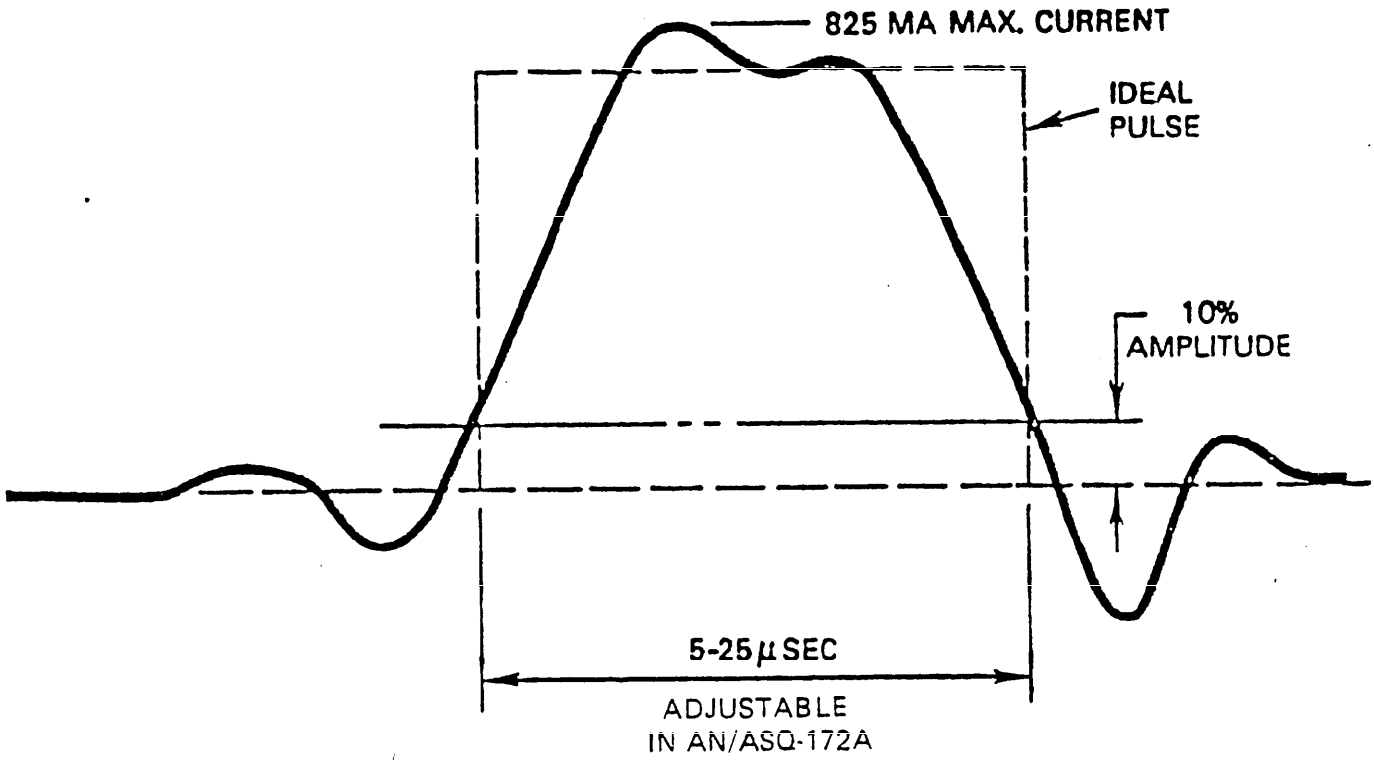


FIGURE 7. KA-99A - AN/ASQ-172A interconnecting cable.

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1. TYPICAL PULSE FIRING INTO LED THROUGH NINE FT. OF CABLE DEFINED IN FIGURE 7.
2. LED FORWARD DROP 4.0 V. MAX AT 750 MA.
3. PULSE SHAPE MAY DEVIATE FROM ABOVE SO LONG AS EQUIVALENT ENERGY IS CONTAINED WITHIN THE REAL ENVELOPE.

FIGURE 8. Typical LED firing pulse.

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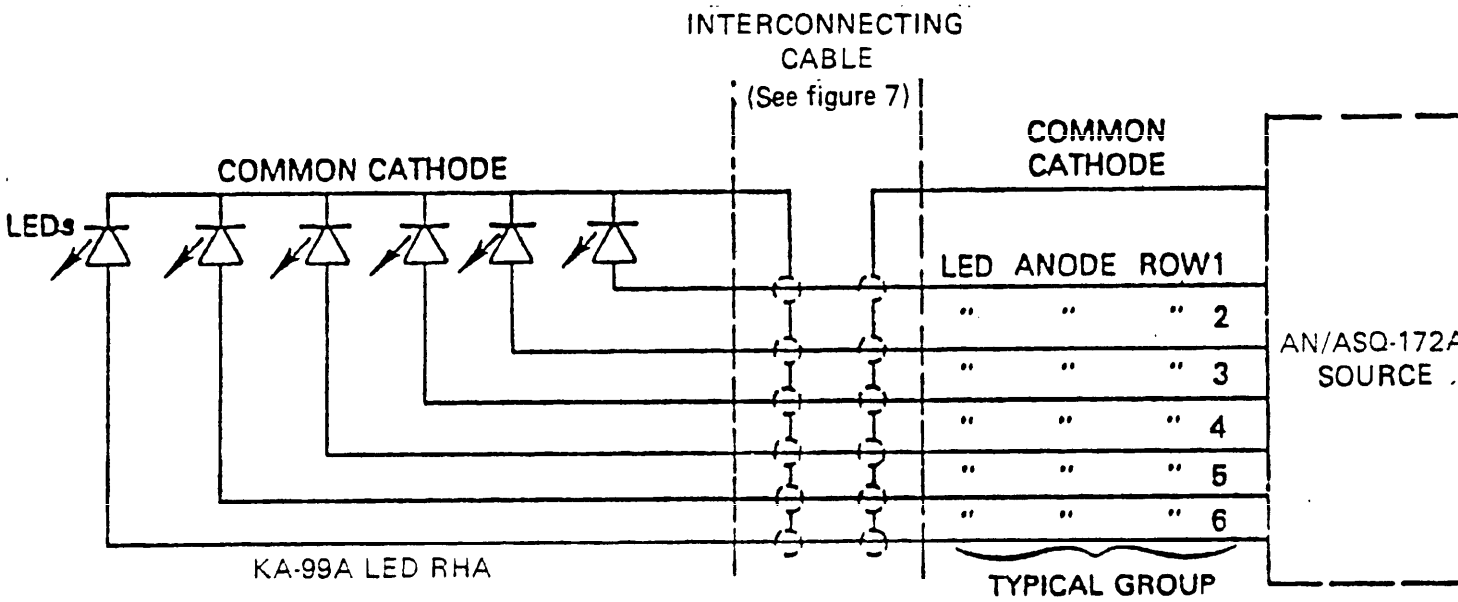
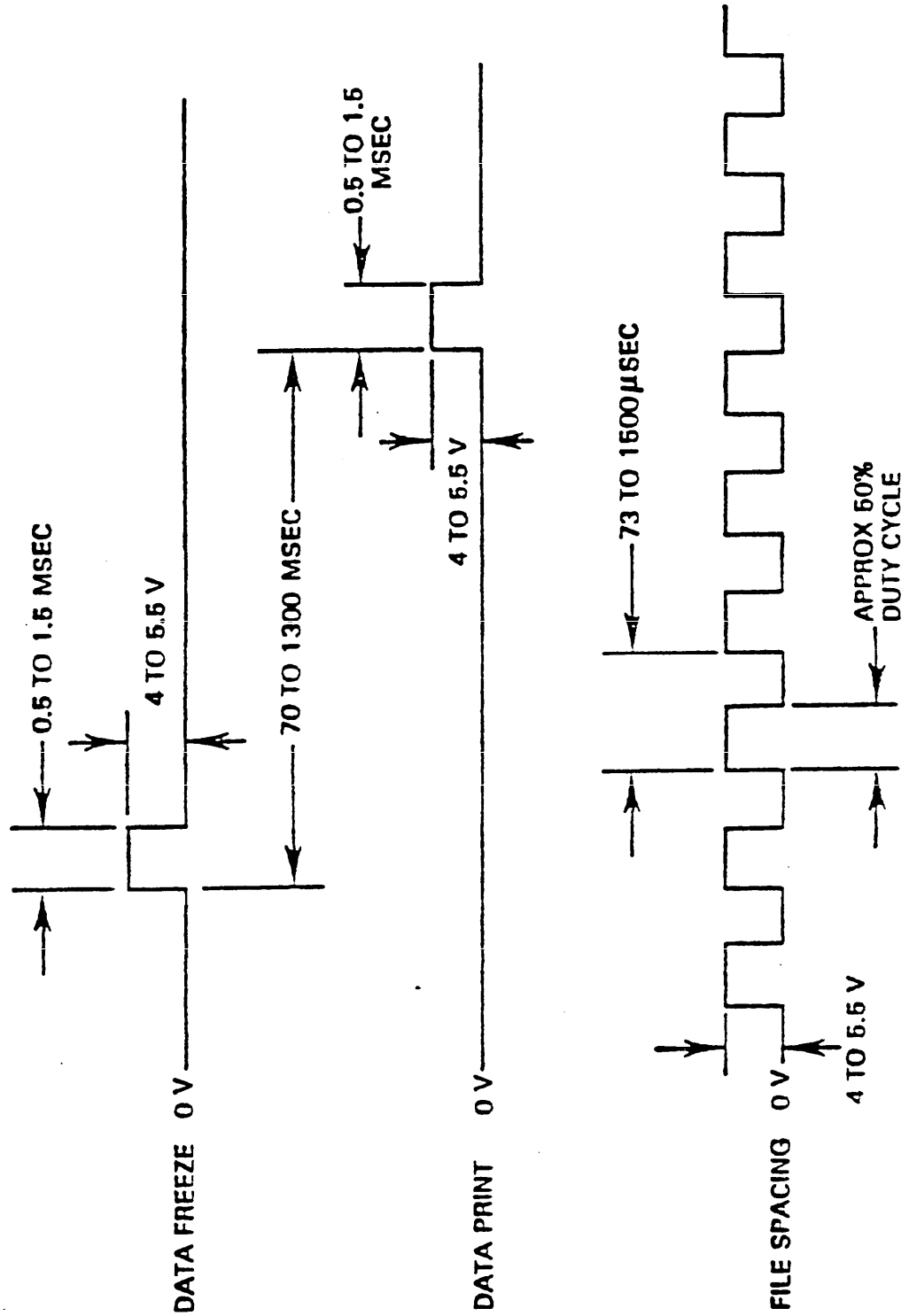


FIGURE 9. LED firing circuit.

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NOTE: WHEN THE KA-99A IS OPERATING, A MINIMUM OF TEN (10) CYCLES OF THE FILE SPACING SIGNAL WILL OCCUR PRIOR TO THE FIRST OCCURRENCE OF DATA FREEZE.

FIGURE 10. KA-99A output timing signals.

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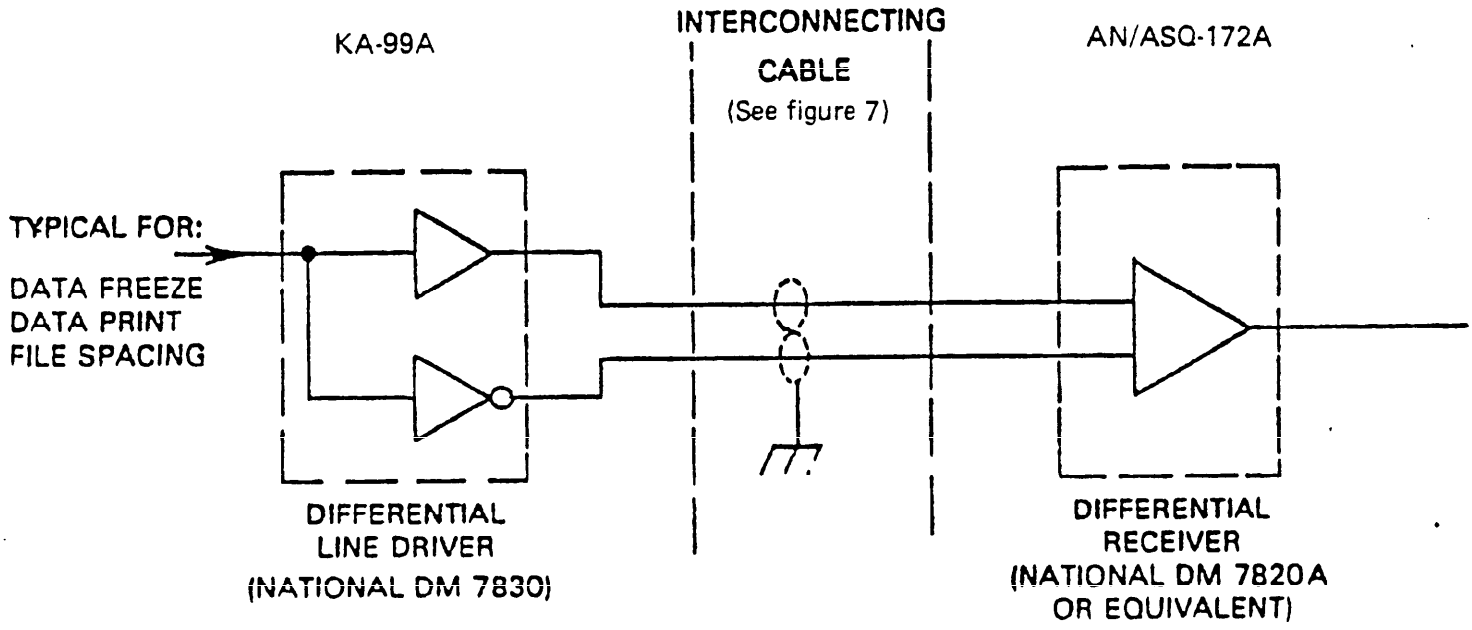


FIGURE 11. KA-99A output timing signal interface circuitry.

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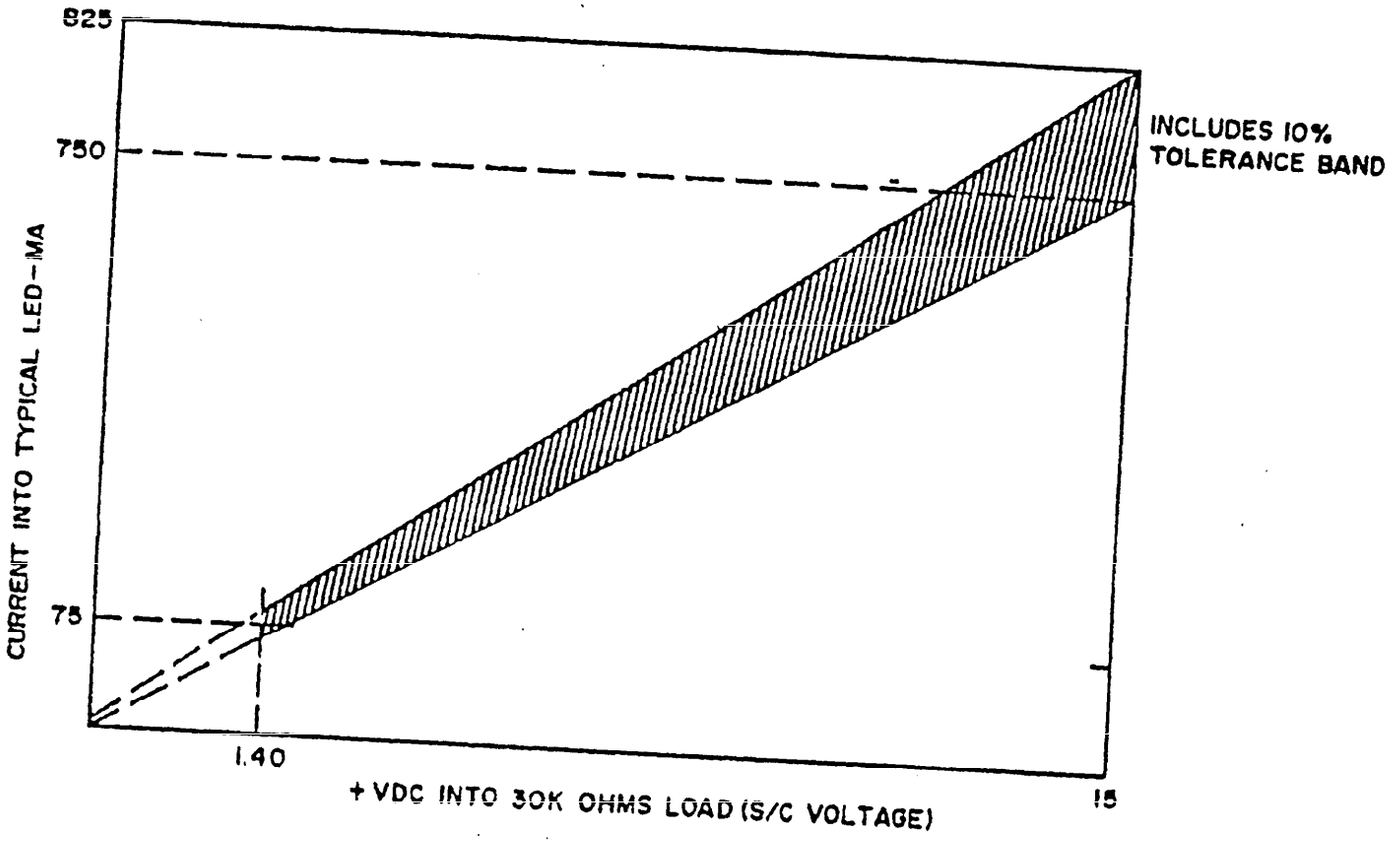


FIGURE 12. Film sensitivity control.

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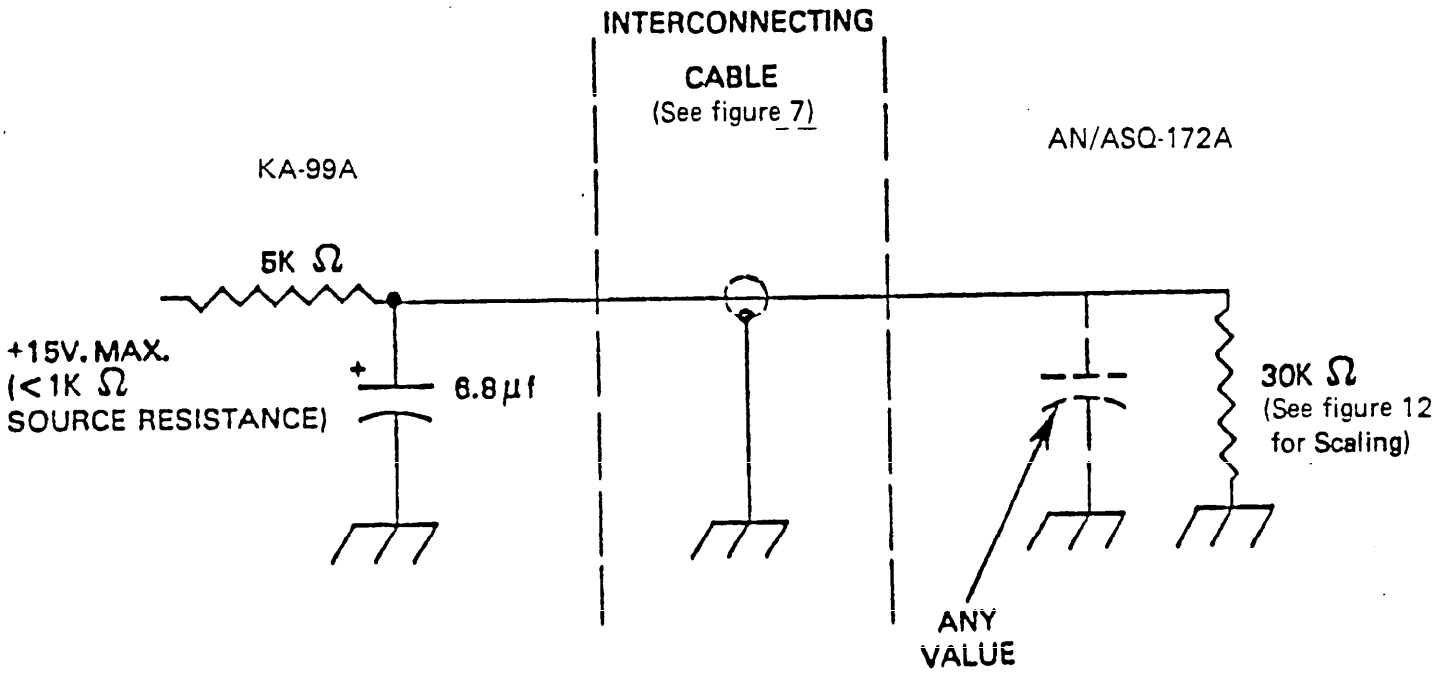
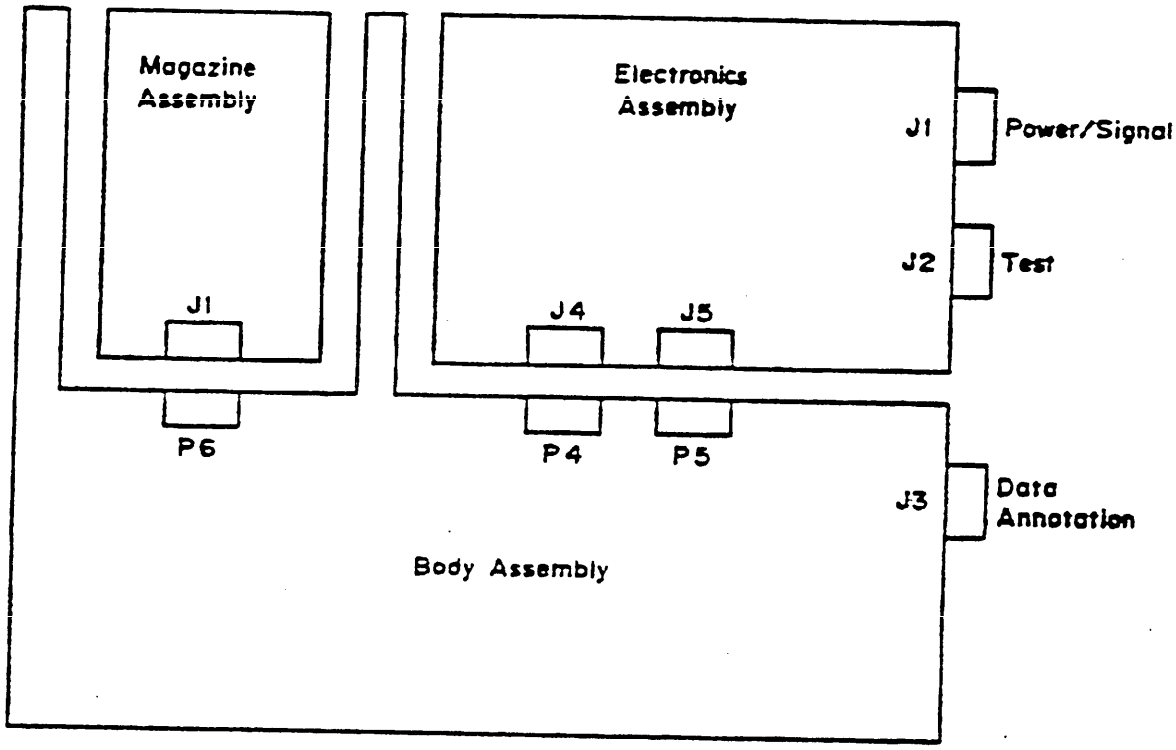


FIGURE 13. Film sensitivity control interface circuitry.

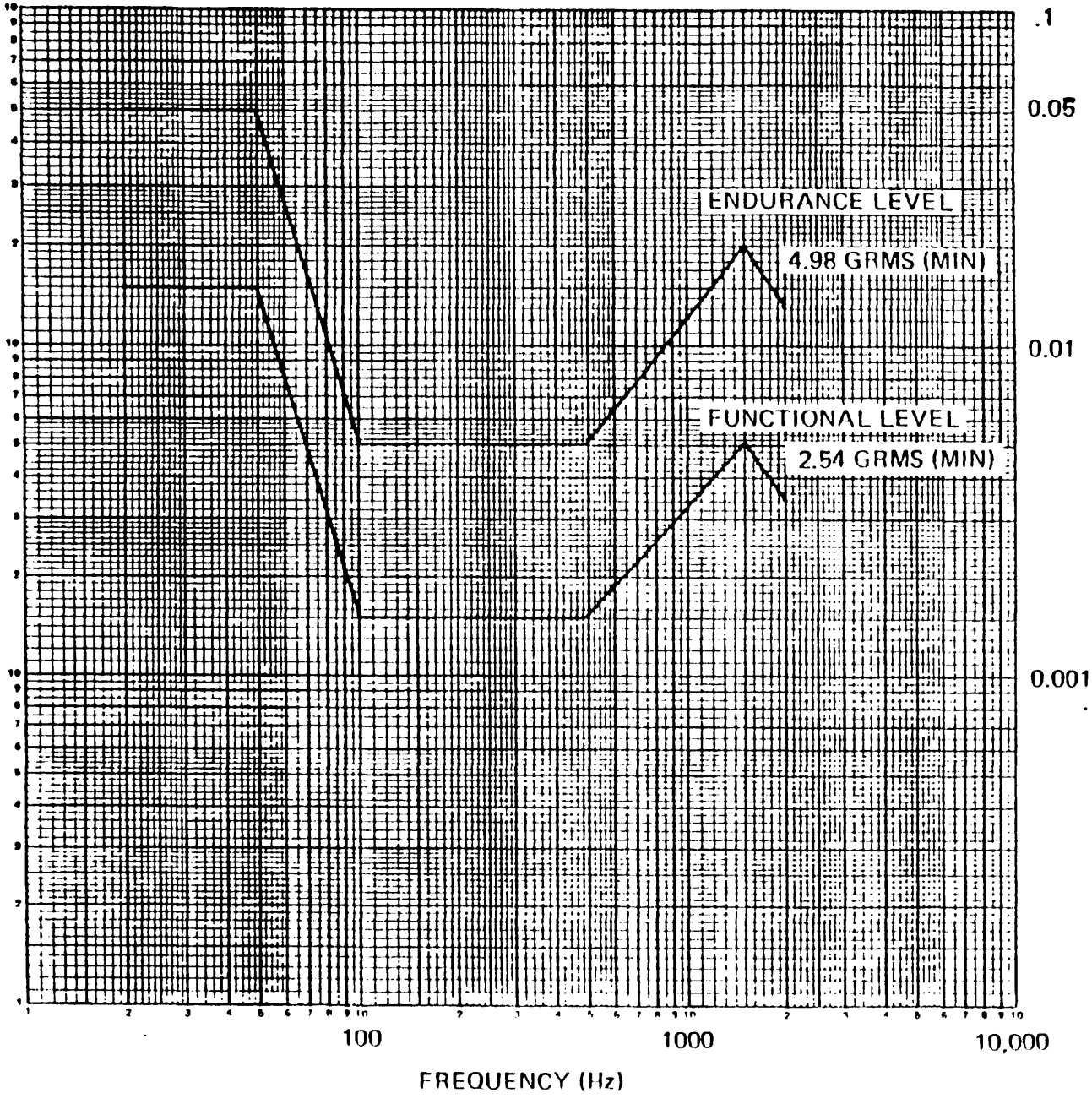
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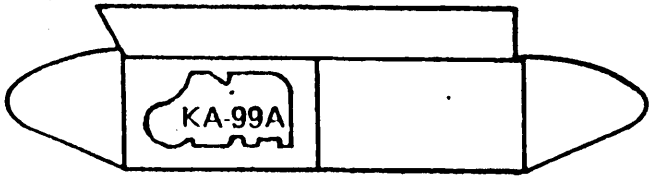


SRA	Ref Design	Connector P/N
Magazine	J1	M24308/4-4
Electronics	J1	MS27656T17B35P
	J2	MS27656T25B35S
	J4,J5	M24308/2-5
Body	J3	MS27656T21B35P
	P4,P5	M24308/4-5
	P6	M24308/2-4

FIGURE 14. Camera system interface connectors.

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FUNCTIONAL LEVEL TEST CURVE DERIVED FROM FLIGHT; THEREFORE, TOLERANCE ON BOTH FUNCTIONAL & ENDURANCE LEVELS ARE: -0 dB
+VENDER'S SYSTEM CAPABILITIES

NOTE: UNIT HARD MOUNTED TO TEST FIXTURE TO INSURE INPUT OF HIGHER FREQUENCIES.

MIL-C-85248A(AS)

FIGURE 15. Vibration qualification curve.

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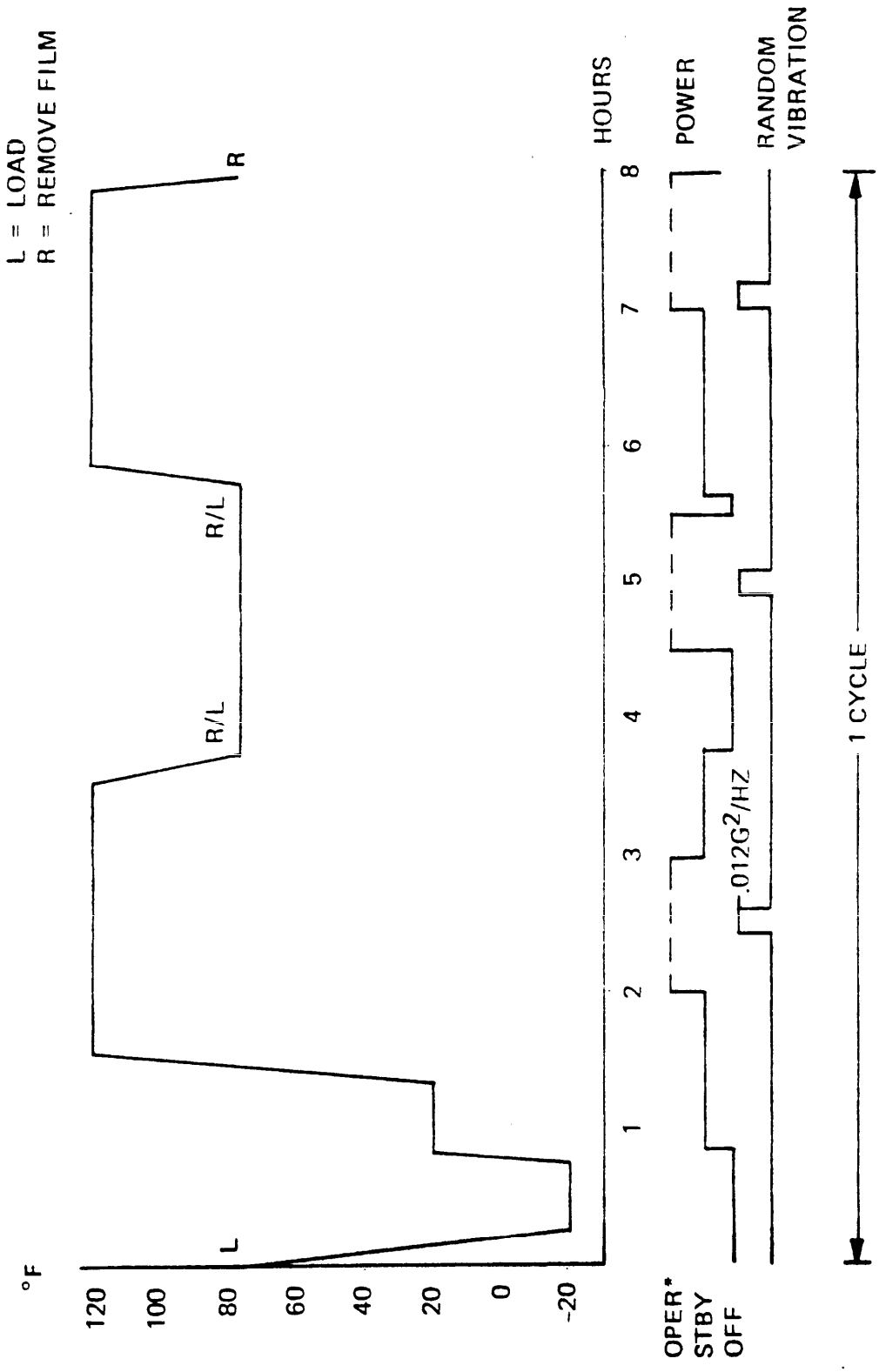


FIGURE 16. Camera reliability demonstration test, environmental cycle

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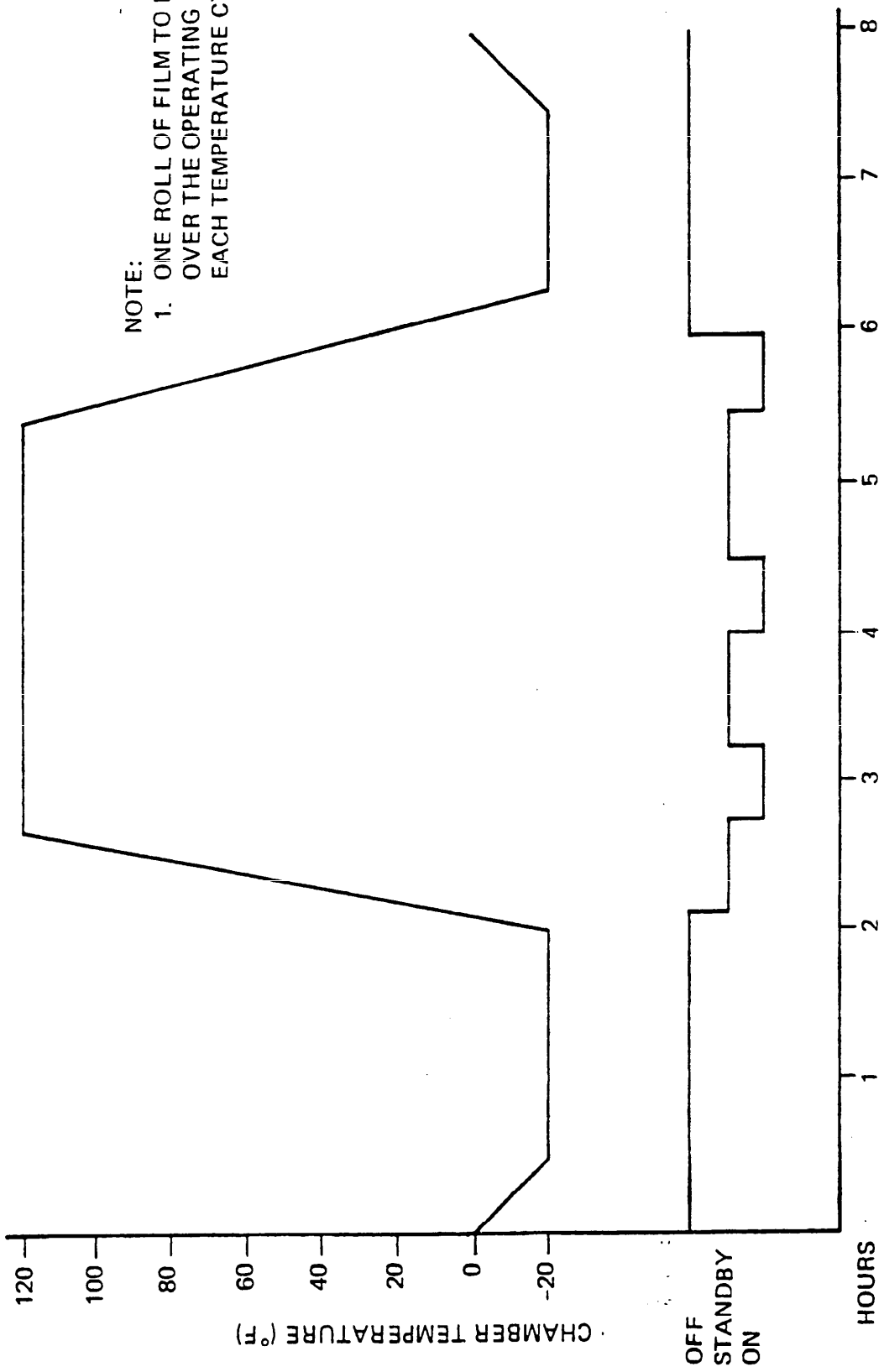


FIGURE 17. Reliability acceptance test profile.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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